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Display apparatus having display module and manufacturing method thereof

Abstract

A display module includes: a substrate including a mounting surface on which a plurality of inorganic light-emitting diodes emitting light in a first direction are mounted, a side surface, and a rear surface being opposite to the mounting surface; a front cover bonded with the mounting surface and covering the mounting surface in the first direction; a metal plate bonded with the rear surface; a side cover surrounding the side surface; and a side end member covering at least one portion of the side cover and extending in a second direction being orthogonal to the first direction along the side surface, wherein the side end member includes a body extending in the second direction, and a plurality of ribs extending from the body in the first direction and arranged at intervals along the second direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a by-pass continuation application of International Application No. PCT/KR2022/021537, filed on Dec. 28, 2022, which is based on and claims priority to Korean Patent Application No. 10-2022-033439, filed on Mar. 17, 2022 in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

1. Field

(1) The disclosure relates to a display apparatus for displaying an image by coupling modules in which self-emissive inorganic light-emitting diodes are mounted on substrates.

2. Description of the Related Art

(2) A display apparatus is a kind of output apparatus for visually displaying images and data information, such as characters, figures, etc.

(3) A display apparatus may include a liquid crystal panel with a backlight or an Organic Light-Emitting Diode (OLED) panel configured with a film of an organic compound that itself emits light in response to current. A liquid crystal panel requires a backlight because itself does not emit light and may have a slow response time and high power consumption. Accordingly, it may be challenging to make a compact liquid crystal panel. An OLED panel requires no backlight and so may be thin. However, an OLED panel may be vulnerable to a burn-in phenomenon in which certain area of the screen remains as it is after the screen changes to another screen due to short life cycles of sub pixels.

(4) A micro light-emitting diode panel (a micro LED panel or μ LED panel) in which inorganic light-emitting diodes are mounted on a substrate and the inorganic light-emitting diodes themselves are used as pixels. The micro light-emitting diode panel may be flat configured with a plurality of inorganic LEDs each having a size of 100 micrometers or less.

(5) The micro LED panel does not cause the burn-in phenomenon of OLEDs as inorganic light-emitting diodes that are self-emissive devices, in addition to having excellent brightness, resolution, consumption power, and durability.

- (6) The micro LED panel provides better contrast, response time, and energy efficiency than the LCD panel requiring the backlight. Micro LEDs that are inorganic light-emitting diodes have higher brightness, higher light-emitting efficiency, and a longer lifespan than OLEDs although both the OLEDs and micro LEDs have high energy efficiency.
- (7) In addition, the micro LEDs may achieve a substrate-level display modulation by arranging LEDs in units of pixels on a circuit board, and provide various resolutions and screen sizes of display according to the customer's order.

SUMMARY

- (8) Provided are a display apparatus, a display module, and a manufacturing method thereof suitable for enlargement and protection of a display apparatus against static electricity.
- (9) According to an aspect of the disclosure, a display module includes: a substrate having a mounting surface on which a plurality of inorganic light-emitting diodes are mounted, a side surface, and a rear surface being opposite to the mounting surface; a front cover bonded with the mounting surface of the substrate and covering the mounting surface in the first direction; a metal plate bonded with the rear surface of the substrate; a side cover surrounding the side surface of the substrate; and a side end member covering at least one portion of the side cover, and extending along the side surface of the substrate in a second direction that is orthogonal to the first direction, wherein the side end member includes a body extending in the second direction, and a plurality of ribs extending from the body in the first direction and arranged at intervals along the second direction.
- (10) The side cover may include: a side end portion forming a side end of the side cover, and a rear portion extending in a rear direction from the side end portion with respect to the first direction and positioned at an inner location than the side end portion in a third direction that is orthogonal to the first direction and the second direction, and the body may be provided on the rear portion in the third direction.
- (11) The plurality of ribs may be provided on the side end portion.
- (12) The plurality of ribs may be provided on the side end portion in the third direction and the rear portion in the third direction.
- (13) The body may be formed at rear ends of the plurality of ribs.
- (14) A length of each of the plurality of ribs in the second direction may be equal to or less than a length of a separation distance between the plurality of ribs in the second direction.
- (15) A conductivity of the side end member may be greater than a conductivity of the side cover.
- (16) A color of the side end member may be a black color.
- (17) The side end member may include a metal material.
- (18) According to an aspect of the disclosure, a display module includes: a substrate having a mounting surface on which a plurality of inorganic light-emitting diodes are mounted, a side surface, and a rear surface being opposite to the mounting surface; a front cover bonded with the mounting surface of the substrate and covering the mounting surface of the substrate; a metal plate bonded with the rear surface of the substrate; a side cover surrounding the side surface of the substrate; and a side end member covering at least one portion of the side cover and at least one portion of the metal plate, the side end member including a first end portion, a second end portion spaced from the first end portion in a direction in which the mounting surface faces, and a side end portion extending from the first end portion to the second end portion and forming a side end; wherein the first end portion of the side end member covers at least one portion of the side end of the side cover, the second end portion of the side end member contacts the metal plate and is grounded to the metal plate, and the side end portion include mountains and valleys alternately arranged in the direction in which the mounting surface faces.
- (19) A mountain from among the mountains may be provided at the first end portion, and a valley among the valleys may be provided at the second end portion.
- (20) A conductivity of the side end member may be greater than a conductivity of the side cover.

- (21) A height of each of the mountains and a depth of each of the valleys are the same.
- (22) The mountains and the valleys may have a same volume.
- (23) According to an aspect of the disclosure, a display apparatus includes: a display module array including a plurality of display modules arranged in a matrix, wherein each of the plurality of display modules includes: a substrate having a mounting surface on which a plurality of inorganic light-emitting diodes are mounted, a side surface, and a rear surface being opposite to the mounting surface; a front cover bonded with the mounting surface of the substrate and covering the mounting surface of the substrate; a metal plate bonded with the rear surface of the substrate; a side cover surrounding the side surface of the substrate; and a side end member covering at least one portion of the side cover and extending along the side surface of the substrate in a second direction that is orthogonal to the first direction, wherein the side end member includes a body extending in the second direction, and a plurality of ribs extending from the body in the first direction and arranged at intervals along the second direction.
- (24) The plurality of display modules may include a first display module and a second display module, and the side end member of the first display module may engage the side end member of the second display module.
- (25) A length of each of the plurality of ribs in the second direction may be equal to or less than a length of a separation distance between the plurality of ribs in the second direction.
- (26) The side cover may include: a side end portion forming a side end of the side cover, and a rear portion extending in a rear direction from the side end portion with respect to the first direction and in a third direction that is orthogonal to the first direction and the second direction, and the body may be provided on the rear portion.
- (27) The plurality of ribs may be provided on the side end portion.
- (28) The plurality of ribs may be provided on the side end portion in the third direction and the rear portion in the third direction.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 shows a display apparatus according to an embodiment of the disclosure;
- (3) FIG. 2 is an exploded view showing main components of the display apparatus of FIG. 1;
- (4) FIG. 3 is an enlarged cross-sectional view showing some components of a display module shown in FIG. 1;
- (5) FIG. 4 is a rear perspective view of the display module of the display apparatus shown in FIG. 1;
- (6) FIG. 5 is a perspective view of some components of the display module shown in FIG. 1;
- (7) FIG. 6 is a cross-sectional view of some components of the display apparatus shown in FIG. 1, taken along a second direction;
- (8) FIG. 7 is an enlarged cross-sectional view of the some components shown in FIG. 6;
- (9) FIG. 8 is a cross-sectional view showing a portion of the display module of the display apparatus of FIG. 1, taken along a third direction;
- (10) FIG. 9 is a cross-sectional view showing some components of the display apparatus of FIG. 1, taken along the third direction;
- (11) FIG. 10 is an enlarged cross-sectional view showing the some components of FIG. 9;
- (12) FIG. 11 is a cross-sectional view showing a portion of a display module of a display apparatus according to another embodiment of the disclosure, taken along the third direction;

- (13) FIG. 12 is a side view showing some components of the display module of the display apparatus of FIG. 11, taken along the third direction;
- (14) FIG. 13 schematically shows a process in which display modules of the display apparatus shown in FIG. 11 are arrayed according to an embodiment of the disclosure;
- (15) FIG. 14 schematically shows a process in which display modules of a display apparatus according to an embodiment of the disclosure are arrayed; and
- (16) FIG. 15 is a cross-sectional view showing selected components of the display modules.

DETAILED DESCRIPTION

- (17) The embodiments described in the present specification are only the most preferred embodiments of the disclosure and do not represent all of the technical idea of the disclosure. It should be understood that various equivalents or variations that may be substituted for them at the time of the present application belong to the scope of rights of the disclosure.
- (18) It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. In the drawings, for ease of understanding, the shapes or sizes of components may be more or less exaggeratedly shown.
- (19) It will be understood that when the terms “includes,” “comprises,” “including,” and/or “comprising,” when used in this specification, specify the presence of stated features, figures, steps, operations, components, members, or combinations thereof, but do not preclude the presence or addition of one or more other features, figures, steps, operations, components, members, or combinations thereof.
- (20) Also, in this specification, the meaning of ‘identical’ may include similar in attribute or similar within a certain range. Also, the term ‘identical’ means ‘substantially identical’. The meaning of ‘substantially identical’ needs to be understood that a value falling within the margin of error in manufacturing or a value corresponding to a difference within a meaningless range with respect to a reference value is included in the range of ‘identical’.
- (21) Throughout the disclosure, the expression “at least one of a, b or c” indicates only a, only b, only c, both a and b, both a and c, both b and c, all of a, b, and c, or variations thereof.
- (22) Hereinafter, embodiments of the disclosure will be described in detail with reference to the accompanying drawings.
- (23) FIG. 1 shows a display apparatus according to an embodiment of the disclosure, FIG. 2 is an exploded view showing main components of the display apparatus of FIG. 1, FIG. 3 is an enlarged cross-sectional view showing some components of a display module shown in FIG. 1, FIG. 4 is a rear perspective view of the display module of the display apparatus shown in FIG. 1, and FIG. 5 is a perspective view of some components of the display module shown in FIG. 1.
- (24) Some components of a display apparatus 1, including a plurality of inorganic light-emitting diodes 50, shown in the drawings may be micro-scale components each having a size of several micrometers (μm) to hundreds of micrometers (μm). For convenience of description, the scale of some components, such as, for example, the plurality of inorganic light-emitting diodes 50, a black matrix 48, etc., may be exaggerated in scale and shown in a larger size in the drawings.
- (25) The display apparatus 1 may be, for example, an apparatus for displaying information, materials, data, etc. in characters, figures, graphs, images, etc., and the display apparatus 1 may be implemented as a television (TV), a personal computer (PC), a mobile device, a digital signage, etc.
- (26) According to an embodiment, as shown in FIGS. 1 and 2, the display apparatus 1 may include a display panel 20 for displaying an image. The example embodiment may include a power supply for supplying power to the display panel 20, a main board 25 for controlling overall operations of the display panel 20, a frame 15 supporting the display panel 20, and a rear cover 10 covering a rear surface of the frame 15.
- (27) The display panel 20 may include a plurality of display modules 30A to 30P, a driving board for driving the display modules 30A to 30P, and a timing controller (TOCN) board for generating timing signals for controlling the display modules 30A to 30P.

(28) The rear cover **10** may support the display panel **20**. The rear cover **10** may be, for example, installed on a floor through a stand or mounted on a wall through a hanger and other installations or arrangements may be used in example embodiments.

(29) The plurality of display modules **30A** to **30P** may be arranged adjacent to one another in up, down, left, and right directions in example embodiments. The plurality of display modules **30A** to **30P** may be arranged in a $M \times N$ matrix form. In the current embodiment, sixteen (16) display modules **30A** to **30P** may be arranged in a 4×4 matrix form. However, the number and arrangement of the plurality of display modules **30A** to **30P** are not limited thereto. Additionally, the various display modules **30A** to **30P** may be placed at different locations from that shown in FIG. 1. For example, the locations of the second display module **30E** and the third display module **30B** may be switched from that shown in FIG. 1.

(30) The plurality of display modules **30A** to **30P** may be mounted on the frame **15**. The plurality of display modules **30A** to **30P** may be mounted on the frame **15** by various known methods, such as, for example, a magnetic force generated by a magnet or a mechanical insert structure. A rear side of the frame **15** may be coupled with the rear cover **10**, and the rear cover **10** may form a rear outer appearance of the display apparatus **1**.

(31) The rear cover **10** may include a metal material. Accordingly, heat generated from the plurality of display modules **30A** to **30P** and the frame **15** may be easily transferred to the rear cover **10**, which raises heat dissipation efficiency of the display apparatus **1**.

(32) As such, the display apparatus **1** according to an embodiment may implement a large screen by tiling the plurality of display modules **30A** to **30P**.

(33) In other embodiments, instead of a matrix including a plurality of display modules **30A** to **30P**, a single display module may be applied to a display apparatus. For example, in an embodiment, a single one of display modules **30A** to **30P** may be installed in and applied to devices such as a wearable device, a portable device, a handheld device, and various electronic products or electronic parts requiring a display. According to other embodiments, a matrix utilizing a plurality of display modules **30A** to **30P** may be assembled and arranged in a matrix and used for these devices. The display modules **30A** to **30P** may be applied to a display apparatus, such as a monitor for PC, a high-resolution TV, a signage, an electronic display, etc. A single display module **30A** to **30P** may be used or a plurality of the modules **30A** to **30P** may be assembled and arranged in a matrix. The plurality of display modules **30A** to **30P** may have the same or similar configuration. Accordingly, the following description about a display module may be applied in the same way to all the other display modules.

(34) Hereinafter, a first display module **30A** of the plurality of display modules **30A** to **30P** will be described.

(35) Components of the plurality of display modules **30A** to **30P**, a display module **30**, a substrate **40**, and a front cover **70** will be representatively described.

(36) Embodiments of the first display module **30A** of the plurality of display modules **30A** to **30P** and a second display module **30E** being adjacent to the first display module **30A** in a second direction Y or a third display module **30B** being adjacent to the first display module **30A** in a third direction Z will also be described as necessary.

(37) Embodiments of a fourth display module **30I** being adjacent to the second display module **30E** in the second direction Y and being opposite to the first display module **30A** with respect to the second display module **30E**, and a fifth display module **30C** being adjacent to the third display module **30B** in the third direction Z and being opposite to the first display module **30A** with respect to the third display module **30B** will also be described as necessary.

(38) The first display module **30A** may be formed in a quadrangle shape. The first display module **30A** may be formed in, for example, a rectangle type or a square shape.

(39) Accordingly, the first display module **30A** may include edges **31**, **32**, **33**, and **34** positioned in upper, lower, left, and right directions with respect to a first direction X which is a front direction.

(40) As shown in FIG. 3, each of the plurality of display modules **30A** to **30P** of the embodiment may include the substrate **40**, and a plurality of inorganic light-emitting diodes **50** mounted on the substrate **40**. The plurality of inorganic light-emitting diodes **50** may be mounted on a mounting surface **41** of the substrate **40** toward the first direction X. In FIG. 3, for convenience of description, a thickness in first direction X of the substrate **40** is shown at an exaggerated scale.

(41) The substrate **40** may be formed in a quadrangle shape. As described above, because each of the plurality of display modules **30A** to **30P** is formed in a quadrangle shape, the substrate **40** may also be formed in a quadrangle shape correspondingly.

(42) The substrate **40** may be formed in a quadrangle shape such as, for example, a square shape or a rectangular shape.

(43) Accordingly, in the example embodiment of the first display module **30A**, the substrate **40** may include four edges E corresponding to the edges **31**, **32**, **33**, and **34** of the first display module **30A**, positioned in the upper, lower, left, and right directions with respect to the first direction X which is the front direction (see FIG. 5).

(44) The substrate **40** may include a substrate body **42**, a mounting surface **41** forming one surface of the substrate body **42**, a rear surface **43** forming the other surface of the substrate body **42** and being opposite to the mounting surface **41**, and a side surface **45** positioned between the mounting surface **41** and the rear surface **43**.

(45) The side surface **45** may form a side end of the substrate **40** in the second direction Y and the third direction Z that are orthogonal to the first direction X.

(46) The substrate **40** may include a chamfer portion **49** formed between the mounting surface **41** and the side surface **45** and between the rear surface **43** and the side surface **45**.

(47) The chamfer portion **49** may prevent, upon an arrangement of the plurality of display modules **30A** to **30P**, each substrate **40** from colliding with another one(s) and being damaged.

(48) The edges E of the substrate **40** may include the side surface **45** and the chamfer portion **49**.

(49) The substrate **40** may include a thin film transistor (TFT) layer **44** formed on the substrate body **42** to drive the inorganic light-emitting diodes **50**. The substrate body **42** may include a glass substrate. That is, the substrate **40** may include a chip on glass (COG) type substrate. On the substrate **40**, a first pad electrode **44a** and a second pad electrode **44b** may be formed to electrically connect the inorganic light-emitting diodes **50** with the TFT layer **44**.

(50) TFTs constituting the TFT layer **44** are not limited to specific structures or types, and may be implemented as various embodiments. TFTs of the TFT layer **44** according to an embodiment may be implemented as, for example, low temperature poly silicon (LTPS) TFTs, oxide TFTs, Si (poly silicon or a-silicon) TFTs, organic TFTs, or graphene TFTs.

(51) Also, the TFT layer **44** may be replaced with complementary metal-oxide semiconductor (CMOS) type, n-type MOSFET, or p-type MOSFET transistors, in a case in which the substrate body **42** of the substrate **40** is a silicon wafer.

(52) The plurality of inorganic light-emitting diodes **50** may be formed of an inorganic material, and each of the inorganic light-emitting diodes **50** may have sizes of several micrometers (μm) to hundreds of micrometers (μm) in width, length, and height. A micro inorganic light-emitting device may have a shorter side length of 100 μm or less of a width, length, or height. The inorganic light-emitting diodes **50** may be picked up from a sapphire or silicon wafer and then directly transferred onto the substrate **40**. The plurality of inorganic light-emitting diodes **50** may be picked up and conveyed through an electrostatic method using an electrostatic head or a stamp method using an elastic polymer material, such as PDMS or silicon, for example, as a head.

(53) Each of the plurality of inorganic light-emitting diodes **50** may be a light-emitting structures including an n-type semiconductor **58a**, an active layer **58c**, a p-type semiconductor **58b**, a first contact electrode **57a**, and a second contact electrode **57b**.

(54) For example, any one of the first contact electrode **57a** or the second contact electrode **57b** may be electrically connected with the n-type semiconductor **58a**, and the other one may be

electrically connected with the p-type semiconductor **58b**.

(55) The first contact electrode **57a** and the second contact electrode **57b** may be flip chip type electrodes arranged horizontally toward the same direction (an opposite direction of a light-emitting direction).

(56) Each inorganic light-emitting device **50** may include a light-emitting surface **54** positioned toward the first direction X upon being mounted on the mounting surface **41**. A side surface **55**, and a bottom surface **56** are opposite to the light-emitting surface **54**. The first contact electrode **57a** and the second contact electrode **57b** may be formed on the bottom surface **56**.

(57) That is, the first and second contact electrodes **57a** and **57b** of the inorganic light-emitting device **50** may be opposite to the light-emitting surface **54**, and accordingly, the first and second contact electrodes **57a** and **57b** may be positioned in the opposite direction of the light-emitting direction.

(58) The first and second contact electrodes **57a** and **57b** may face the mounting surface **41**, and be electrically connected with the TFT layer **44**. Also, the light-emitting surface **54** through which light is irradiated may be positioned in an opposite direction of the direction in which the first and second contact electrodes **57a** and **57b** are positioned.

(59) Accordingly, light generated by the active layer **58c** may be irradiated toward the first direction X through the light-emitting surface **54**, without any interference by the first or second contact electrode **57a** or **57b**.

(60) That is, the first direction X may be defined as a direction in which the light-emitting surface **54** is positioned to irradiate light.

(61) The first contact electrode **57a** and the second contact electrode **57b** may be electrically connected respectively with the first pad electrode **44a** and the second pad electrode **44b** formed on the mounting surface **41** of the substrate **40**.

(62) The inorganic light-emitting device **50** may be connected directly with the first and second pad electrodes **44a** and **44b** through an anisotropic conductive layer **47** or a bonding material such as a solder.

(63) On the substrate **40**, the anisotropic conductive layer **47** may be formed to mediate an electrical connection between the first and second contact electrodes **57a** and **57b** and the first and second pad electrodes **44a** and **44b**. The anisotropic conductive layer **47** may be formed by applying an anisotropic conductive adhesive on a protective film, and have a structure in which conductive balls **47a** are distributed in an adhesive resin. Each conductive ball **47a** may be a conductive sphere surrounded by a thin insulating film, and as a result of breaking of the insulating film by pressure, the conductive ball **47a** may electrically connect a conductor with another one.

(64) The anisotropic conductive layer **47** may include an anisotropic conductive film (ACF) being in a form of a film, and an anisotropic conductive paste (ACP) being in a form of a paste.

(65) In an embodiment, the anisotropic conductive layer **47** may be provided as an anisotropic conductive film.

(66) The insulating films of the conductive balls **47a** may be broken by pressure applied to the anisotropic conductive layer **47** upon mounting of the plurality of inorganic light-emitting diodes **50** on the substrate **40**. As a result, the first and second contact electrodes **57a** and **57b** of the inorganic light-emitting diodes **50** may be electrically connected with the first and second pad electrodes **44a** and **44b** of the substrate **40**.

(67) In some embodiments, the plurality of inorganic light-emitting diodes **50** may be, for example, mounted on the substrate **40** through a solder, instead of the anisotropic conductive layer **47**. By performing a reflow process after arranging the inorganic light-emitting diodes **50** on the substrate **40**, the inorganic light-emitting diodes **50** may be bonded on the substrate **40**.

(68) The plurality of inorganic light-emitting diodes **50** may include a red light-emitting device **51**, a green light-emitting device **52**, and a blue light-emitting device **53**. The inorganic light-emitting diodes **50** may be mounted in unit of a group including a series of a red light-emitting device **51**, a

green light-emitting device **52**, and a blue light-emitting device **53** on the mounting surface **41** of the substrate **40**. The series of the red light-emitting device **51**, the green light-emitting device **52**, and the blue light-emitting device **53** may form a pixel. In this case, each of the red light-emitting device **51**, the green light-emitting device **52**, and the blue light-emitting device **53** may form a sub pixel.

(69) The red light-emitting device **51**, the green light-emitting device **52**, and the blue light-emitting device **53** may be aligned with preset intervals, as in the current embodiment. In other embodiments, the red light-emitting device **51**, the green light-emitting device **52**, and the blue light-emitting device **53** may be arranged in other arrangements such as, for example, a triangular arrangement.

(70) The substrate **40** may include a light absorbing layer **44c** for absorbing external light to improve contrast. The light absorbing layer **44c** may be formed on the entire of the mounting surface **41** of the substrate **40**. The light absorbing layer **44c** may be formed between the TFT layer **44** and the anisotropic conductive layer **47**.

(71) The plurality of display modules **30A** to **30P** may further include the black matrix **48** formed between the plurality of inorganic light-emitting diodes **50**.

(72) The black matrix **48** may function to supplement the light absorbing layer **44c** formed on the entire of the mounting surface **41** of the substrate **40**. That is, the black matrix **48** may improve contrast of a screen by absorbing external light such that the substrate **40** is shown to be black.

(73) The black matrix **48** may have a black color.

(74) According to the current embodiment, the black matrix **48** may be positioned between pixels each formed by a series of a red light-emitting device **51**, a green light-emitting device **52**, and a blue light-emitting device **53**. In other embodiments, the black matrix **48** may be formed more finely to partition the red light-emitting device **51**, the green light-emitting device **52**, and the blue light-emitting device **53** that are sub pixels.

(75) The black matrix **48** may be formed in a shape of a lattice having a horizontal pattern and a vertical pattern to be positioned between the pixels.

(76) The black matrix **48** may be formed, for example, by applying a light absorbing ink on the anisotropic conductive layer **47** through an ink-jet process and then hardening the light absorbing ink or by coating the anisotropic conductive layer **47** with a light absorbing film.

(77) That is, in embodiments, the black matrix **48** may be formed in areas between the plurality of inorganic light-emitting diodes **50**, in which none of the plurality of inorganic light-emitting diodes **50** is mounted, on the anisotropic conductive layer **47** formed on the entire of the mounting surface **41**.

(78) Each of the plurality of display modules **30A** to **30P** may include a front cover **70** positioned in the front direction X on the mounting surface **41** of the display module **30A** to **30P** to cover the mounting surface **41**.

(79) A plurality of front covers **70** may be respectively formed on the plurality of display modules **30A** to **30P** in the first direction X (see FIGS. **6** and **7**).

(80) The plurality of display modules **30A** to **30P** may be assembled with each other after the front covers **70** are respectively formed on the display modules **30A** to **30P**. That is, in an example of the first display module **30A** and the second display module **30E** among the plurality of display modules **30A** to **30P**, a first front cover **70A** may be formed on the mounting surface **41** of the first display module **30A** and a second front cover **70B** may be formed on the mounting surface **41** of the second display module **30E**.

(81) The front cover **70** may cover the substrate **40** to protect the substrate **40** against an external force or outside water.

(82) A plurality of layers (not shown) of the front cover **70** may be provided as functional films having optical performance. The plurality of layers will be described in detail, below.

(83) A part of the plurality of layers of the front cover **70** may include a base layer (not shown)

formed of an optical clear resin (OCR). The base layer (not shown) may support the other layers (not shown). The OCR may be in a very transparent state as the OCR may have a transmittance of 90% or more.

(84) The OCR may improve visibility and image quality by raising transmittance through a low reflection property. That is, in a structure having an air gap, light loss occurs due to a refractive index difference between a film layer and an air layer. However, in a structure having an OCR, such a refractive index difference may be reduced to decrease light loss, resulting in an improvement of visibility and image quality.

(85) That is, the OCR may improve image quality, in addition to protecting the substrate **40**.

(86) For example, a part of the plurality of layers of the front cover **70** may include an adhesive layer for bonding the front cover **70** with the mounting surface **41** of the substrate **40**.

(87) The front cover **70** may have a height that is greater than or equal to a preset height in the first direction X in which the mounting surface **41** or the light-emitting surface **54** faces.

(88) The front cover **70** formed on the substrate **40** may have a height capable of sufficiently filling a gap that may be formed between the plurality of inorganic light-emitting diodes **50** with respect to the front cover **70**.

(89) Each of the plurality of display modules **30A** to **30P** may include a metal plate **60** positioned on a rear surface **43** of the substrate **40**.

(90) Also, each of the plurality of display modules **30A** to **30P** may include a rear adhesive tape **61** positioned between the rear surface **43** of the substrate **40** and the metal plate **60** to bond the rear surface **43** with the metal plate **60**.

(91) The rear adhesive tape **61** may be a double-sided adhesive tape. However, the rear adhesive tape **61** is not limited to a double-sided adhesive tape. Also, in some embodiments, an adhesive, bonding agent or other material, structure or method for bonding the metal plate **60** with the rear surface **43** of the substrate **40** may be used in place of or additionally to the rear adhesive tape **61**. For example, an adhesive layer that is not a tape may be used in some embodiments instead of or in addition to the rear adhesive tape **61**. The plurality of inorganic light-emitting diodes **50** may be electrically connected with a pixel driving wiring) formed on the mounting surface **41**, and an upper wiring layer extending through the side surface **45** of the substrate **40** and formed with a pixel driving wiring.

(92) The upper wiring layer may be formed below the anisotropic conductive layer **47**. The upper wiring layer may be electrically connected with a side wiring **46** formed on the side surface **45** of the substrate **40**. The side wiring **46** may be provided in a form of a thin film. The side wiring **46** may include a coating member **46a** surrounding the side wiring **46** to prevent the side wiring **46** from being exposed to outside and damaged (see FIG. 8).

(93) A left-right direction of the display apparatus **1**, which is orthogonal to the first direction X being a front direction from the display apparatus **1**, is a second direction Y. An up-down direction of the display apparatus **1**, which is orthogonal to the first direction X and the second direction Y, is a third direction Z. The side wiring **46** may extend to the rear surface **43** of the substrate **40** in the third direction Z along the chamfer portion **49** and the side surface **45** of the substrate **40** extending in the third direction Z, although its direction of extension is not limited thereto. For example, the side wiring **46** may extend to the rear surface **43** of the substrate **40** in the second direction Y along the chamfer portion **49** and the side surface **45** of the substrate **40** extending in the second direction Y.

(94) According to an embodiment, the side wiring **46** may extend along edges E of the substrate **40**, corresponding to an upper edge **32** and a lower edge **34** of the first display module **30A**, although not limited thereto.

(95) For example, the side wiring **46** may extend along edges E of the substrate **40**, corresponding to at least two edges of the four edges **31**, **32**, **33**, and **34** of the first display module **30A**.

(96) The upper wiring layer may be connected with the side wiring **46** by a front connecting pad

formed on the edges E of the mounting surface **41**.

(97) The side wiring **46** may extend along the side surface **45** of the substrate **40** and be connected with a rear wiring layer **43b** formed on the rear surface **43**.

(98) An insulating layer **43c** may be formed on the rear wiring layer **43b** in a direction which the rear surface **43** of the substrate **40** faces, to cover the rear wiring layer **43b**.

(99) That is, the plurality of inorganic light-emitting diodes **50** may be electrically connected with the upper wiring layer, the side wiring **46**, and the rear wiring layer **43b**, sequentially.

(100) Also, as shown in FIG. **4**, the first display module **30A** may include a driving circuit board **80** for electrically controlling the plurality of inorganic light-emitting diodes **50** mounted on the mounting surface **41**. The driving circuit board **80** may be a printed circuit board. The driving circuit board **80** may be positioned on the rear surface **43** of the substrate **40** in the first direction X. The driving circuit board **80** may be positioned on the metal plate **60** bonded on the rear surface **43** of the substrate **40**.

(101) The first display module **30A** may include a flexible film **81** connecting the driving circuit board **80** with the rear wiring layer **43b** to electrically connect the driving circuit board **80** with the plurality of inorganic light-emitting diodes **50**.

(102) More specifically, one end of the flexible film **81** may be connected with a rear connecting pad **43d** positioned on the rear surface **43** of the substrate **40** and electrically connected with the plurality of inorganic light-emitting diodes **50**.

(103) The rear connecting pad **43d** may be electrically connected with the rear wiring layer **43b**. Accordingly, the rear connecting pad **43d** may electrically connect the rear wiring layer **43b** with the flexible film **81**.

(104) Because the flexible film **81** is electrically connected with the rear connecting pad **43d**, the flexible film **81** may transfer power and an electrical signal from the driving circuit board **80** to the plurality of inorganic light-emitting diodes **50**. The flexible film **81** may be a flexible flat cable (FFC) or a chip on film (COF). The flexible film **81** may include a first flexible film **81a** and a second flexible film **81b** respectively positioned in the upper and lower directions with respect to the first direction X which is the front direction, although not limited thereto. For example, the first and second flexible films **81a** and **81b** may be positioned in the left and right directions with respect to the first direction X, or in at least two directions of the up, down, left, and right directions with respect to the first direction X.

(105) A plurality of second flexible films **81b** may be provided, although not limited thereto. However, a single second flexible film **81b** may be provided, and a plurality of first flexible films **81a** may also be provided.

(106) The first flexible film **81a** may transfer a data signal from the driving circuit board **80** to the substrate **40**. The first flexible film **81a** may be a COF.

(107) The second flexible film **81b** may transfer power from the driving circuit board **80** to the substrate **40**. The second flexible film **81b** may be a FFC.

(108) For example, the first flexible film **81a** and the second flexible film **81b** may be a FFC and a COF, respectively.

(109) The driving circuit board **80** may be electrically connected with the main board **25** (see FIG. **2**), although not shown in the drawings. The main board **25** may be positioned behind the frame **15**, and the main board **25** may be connected with the driving circuit board **80** through a cable behind the frame **15**.

(110) A fixing member **82** for bonding the display modules **30A** to **30P** to the frame **15** may be positioned on a rear surface of the metal plate **60**. The fixing member **82** may be, for example, a double-sided tape. The metal plate **60** forming a rear side of each of the display modules **30A** to **30P** may be bonded directly to the frame **15** by the fixing member **82**, and accordingly, the display modules **30A** to **30P** may be supported by the frame **15**.

(111) As described above, the metal plate **60** may be in contact with the substrate **40**. The metal

plate **60** may be bonded with the substrate **40** by the rear adhesive tape **61** positioned between the rear surface **43** of the substrate **40** and the metal plate **60**.

(112) FIG. **5** shows the substrate **40** with selected components. Some components may not be illustrated in FIG. **5**, for convenience of description. For example, the anisotropic conductive layer **47** is excluded, for convenience of description. Also, although the side wiring **46** includes the coating member **46a** for protecting the side wiring **46** from outside, the coating member **46a** is omitted for convenience of description.

(113) The metal plate **60** may be formed of a metal material having high thermal conductivity. For example, the metal plate **60** may be formed of an aluminum material.

(114) Heat generated by the TFT layer **44** and the plurality of inorganic light-emitting diodes **50** mounted on the substrate **40** may be transferred to the metal plate **60** through the rear adhesive tape **61** along the rear surface **43** of the substrate **40**.

(115) Accordingly, heat generated by the substrate **40** may be easily transferred to the metal plate **60**, and temperature of the substrate **40** may be prevented from rising to certain temperature or higher.

(116) The plurality of display modules **30A** to **30P** may be arranged at various locations in a $M \times N$ matrix form. The display modules **30A** to **30P** may be independently movable. In this case, because each of the display modules **30A** to **30P** includes the metal plate **60**, a constant level of heat dissipation performance may be maintained regardless of the locations of the display modules **30A** to **30P**.

(117) The display apparatus **1** may form screens having various sizes by arranging the display modules **30A** to **30P** in various forms of $M \times N$ matrixes. Therefore, by including the metal plate **60** in each of the display modules **30A** to **30P** to cause each of the display modules **30A** to **30P** to dissipate heat independently, according to an embodiment, instead of dissipating heat through a single metal plate for temporary heat dissipation, total heat dissipation performance of the display apparatus **1** may be improved.

(118) In a case in which a single metal plate is positioned inside the display apparatus **1**, the metal plate may be not positioned at locations corresponding to some display modules with respect to a front-rear direction, while being positioned at locations where no display modules are positioned, which lowers heat dissipation efficiency of the display apparatus **1**.

(119) That is, by positioning the metal plate **60** in each of the display modules **30A** to **30P** to cause each of the display modules **30A** to **30P** to dissipate heat at any location through the metal plate **60**, the total heat dissipation performance of the display apparatus **1** may be improved.

(120) The metal plate **60** may be provided in a shape of a rectangle substantially corresponding to a shape of the substrate **40**.

(121) An area of the substrate **40** may be at least equal to or larger than an area of the metal plate **60**. Upon a parallel arrangement of the substrate **40** and the metal plate **60** in the first direction X , the four edges E of the substrate **40** being in the shape of the rectangle may correspond to four edges of the metal plate **60** with respect to a center of the substrate **40** and the metal plate **60**, or the four edges E of the substrate **40** may be positioned at outer locations than the four edges of the metal plate **60** with respect to the center of the substrate **40** and the metal plate **60**.

(122) The four edges E of the substrate **40** may be positioned at more outward locations than the four edges of the metal plate **60**. That is, the area of the substrate **40** may be larger than the area of the metal plate **60**.

(123) Upon transferring of heat to the display modules **30A** to **30P**, the substrate **40** and the metal plate **60** may be heat-expanded. Because the metal plate **60** has a greater coefficient of thermal expansion than the substrate **40**, a degree of expansion of the metal plate **60** may be greater than a degree of expansion of the substrate **40**.

(124) In a case in which the four edges E of the substrate **40** correspond to the four edges of the metal plate **60** or are positioned at inner locations than the four edges of the metal plate **60**, the

edges of the metal plate **60** may protrude to outside of the substrate **40**.

(125) Accordingly, separation distances of gaps formed between the display modules **30A** to **30P** may become irregular due to heat-expansion of the metal plate **60** of each of the display modules **30A** to **30P**, and visibility of some seams may rise, which may result in a deterioration of a performance of a screen of the display panel **20**.

(126) However, by positioning the four edges **E** of the substrate **40** at more outward locations than the four edges of the metal plate **60**, the metal plate **60** may not protrude to the outside of the four edges **E** of the substrate **40** even if the substrate **40** and the metal plate **60** are heat-expanded. Accordingly, the gaps formed between the display modules **30A** to **30P** may be maintained with regular separation distances.

(127) In addition, to maintain constant separation distances of gaps formed between the display modules **30A** to **30P**, the frame **15** supporting the display modules **30A** to **30P** may include a front surface having a similar material property to that of the substrate **40**. That is, the display modules **30A** to **30P** may be bonded on the front surface of the frame **15**.

(128) According to an embodiment, the area of the substrate **40** may substantially correspond to the area of the metal plate **60**. Accordingly, heat generated from the substrate **40** may be uniformly dissipated over the entire area of the substrate **40** without being isolated in some areas.

(129) The metal plate **60** may be bonded on the rear surface **43** of the substrate **40** by the rear adhesive tape **61**.

(130) The rear adhesive tape **61** may have a size corresponding to the metal plate **60**. That is, an area of the rear adhesive tape **61** may correspond to the area of the metal plate **60**. The metal plate **60** may be substantially in a shape of a rectangle, and the rear adhesive tape **61** may also be in a shape of a rectangle correspondingly.

(131) The edges of the metal plate **60** being in the shape of the rectangle may correspond to the edges of the rear adhesive tape **61** being in the shape of the rectangle with respect to a center of the metal plate **60** and the rear adhesive tape **61**.

(132) Accordingly, the metal plate **60** and the rear adhesive tape **61** may be easily manufactured as a coupled configuration, thereby increasing manufacturing efficiency of the display apparatus **1**.

(133) That is, before a plate is cut into unit pieces to form the metal plate **60**, the rear adhesive tape **61** may be bonded on the plate and then the rear adhesive tape **61** and the plate may be cut together into unit pieces to form the metal plate **60**, thereby reducing a number of processes.

(134) Heat generated from the substrate **40** may be transferred to the metal plate **60** through the rear adhesive tape **61**. Accordingly, the rear adhesive tape **61** may bond the metal plate **60** on the substrate **40**, while transferring heat generated from the substrate **40** to the metal plate **60**.

(135) Accordingly, the rear adhesive tape **61** may include a material having high heat dissipation performance.

(136) The rear adhesive tape **61** may include, basically, a material having an adhesive property to bond the substrate **40** with the metal plate **60**.

(137) Additionally, the rear adhesive tape **61** may include a material having high heat dissipation performance rather than materials having an adhesive property. Accordingly, the rear adhesive tape **61** may efficiently transfer heat between the substrate **40** and the metal plate **60**.

(138) Also, the material having the adhesive property, included in the rear adhesive tape **61**, may be a material having higher heat dissipation performance than adhesive materials constituting existing adhesives.

(139) The material having the higher heat dissipation performance may be a material capable of effectively transferring heat because the material has high heat conductivity, high heat transfer performance, and low specific heat.

(140) For example, the rear adhesive tape **61** may include a graphite material, although not limited thereto. However, the rear adhesive tape **61** may be formed of any material having high heat dissipation performance.

(141) Flexibility of the rear adhesive tape **61** may be greater than flexibility of the substrate **40** and the metal plate **60**. Accordingly, the rear adhesive tape **61** may be formed of a material having an adhesive property, heat dissipation, and high flexibility. The rear adhesive tape **61** may be a baseless double-sided tape. In this case, the rear adhesive tape **61** may be formed as a single layer of which one side is bonded on the substrate **40** and the other side is bonded on the metal plate **60**, without having any base supporting the one side and the other side.

(142) Because the rear adhesive tape **61** may include no base, the rear adhesive tape **61** may include no material interfering with heat conduction, and accordingly, heat dissipation performance may increase. However, the rear adhesive tape **61** is not limited to a baseless double-sided tape, and in some embodiments may be a heat dissipation tape having higher heat dissipation performance than existing double-sided tapes.

(143) The rear adhesive tape **61** may be formed of a material having high flexibility to absorb an external force transferred from the substrate **40** and the metal plate **60**. More specifically, flexibility of the rear adhesive tape **61** may be higher than flexibility of the substrate **40** and the metal plate **60**.

(144) Accordingly, upon transferring of an external force generated by changed sizes of the substrate **40** and the metal plate **60** by heat transferred to the substrate **40** and the metal plate **60** to the rear adhesive tape **61**, the rear adhesive tape **61** may be deformed to prevent the external force from being transferred to the other components.

(145) The rear adhesive tape **61** may have a certain thickness in the first direction X. The metal plate **60** may be heat-expanded by heat or cooled to be contracted. In this case, the metal plate **60** may be expanded or contracted in the first direction X and directions that are orthogonal to the first direction X, and accordingly, an external force may be transferred to the substrate **40**.

(146) As described above, because the metal plate **60** is formed with a size corresponding to the substrate **40** and covers the entire of the rear surface **43** of the substrate **40**, a fixing member **82** may be positioned on a rear surface of the metal plate **60**, although embodiments are not limited thereto. For example, the fixing member **82** may be positioned on the rear surface **43** of the substrate **40**. In this case, the substrate **40** may be bonded directly on the frame **15** through the fixing member **82**.

(147) In other embodiments, the metal plate **60** may cover only a portion of the rear surface **43** of the substrate **40**, and the fixing member **82** may be bonded on an area not covered by the metal plate **60** in the rear surface **43** of the substrate **40**.

(148) The fixing member **82** may be a double-sided tape.

(149) Hereinafter, the front cover **70**, a side cover **90**, and a side end member **100** will be described in detail.

(150) FIG. **6** is a cross-sectional view of some components of the display apparatus shown in FIG. **1**, taken along a second direction, FIG. **7** is an enlarged cross-sectional view of the some components shown in FIG. **6**, FIG. **8** is a cross-sectional view showing a portion of the display module of the display apparatus of FIG. **1**, taken along a third direction, FIG. **9** is a cross-sectional view showing some components of the display apparatus of FIG. **1**, taken along the third direction, and FIG. **10** is an enlarged cross-sectional view showing the some components of FIG. **9**.

(151) The front cover **70** may protect the substrate **40** from an external force, reduce the visibility of a seam caused by a gap G formed between the plurality of display modules **30A** to **30P**, and improve color deviation between the plurality of display modules **30A** to **30P**.

(152) The plurality of display modules **30A** to **30P** may include the side cover **90** positioned in the gap G formed between the plurality of display modules **30A** to **30P** upon arraying of the plurality of display modules **30A** to **30P**.

(153) To absorb light reflected in the gap G between the plurality of display modules **30A** to **30P**, the front cover **70** of each of the display modules **30A** to **30P** may extend to the outside of the substrate **40** of the display module **30A** to **30P**. A side end **75** of the front cover **70** may extend up

to the outside of the mounting surface **41**.

(154) More specifically, the front cover **70** may extend up to an outer location than an edge (or referred to as a side end) **41S** of the mounting surface **41** of the substrate **40** in the second direction Y and the third direction Z.

(155) A gap G between the display modules **30A** to **30P** may be formed between the side surfaces **45** of the substrates **40** of the display modules **30A** to **30P**. The gap G according to an embodiment may represent a non-display area that may be generated between the display modules **30A** to **30P**, and therefore, the gap G formed between the plurality of display modules **30A** to **30P** may be a space formed between the edges **41S** of the mounting surfaces **41** of the substrates **40** of neighboring ones of the display modules **30A** to **30P**.

(156) Accordingly, the gap G formed between the plurality of display modules **30A** to **30P** may represent a space formed between the edges **41S** of the mounting surfaces **41** of the display modules **30A** to **30P** being adjacent to each other in the second direction Y or the third direction Z.

(157) The front cover **70** extending from the display modules **30A** to **30P** may be positioned in the gap G between the display modules **30A** to **30P** to absorb light irradiated toward the gap G or light reflected in the gap G, thereby reducing visibility of a seam.

(158) Also, the side cover **90** of the display modules **30A** to **30P**, positioned in the gap G, may absorb light irradiated toward the gap G, thereby reducing visibility of a seam, which will be described below.

(159) As shown in FIGS. **6** and **7**, the front cover **70** may extend to the outside of the substrate **40** in the second direction Y. More specifically, the front cover **70** may extend up to an outer location than the side surface **45** and the chamfer portion **49** in the second direction Y.

(160) According to an embodiment, a description about only an edge of the substrate **40** corresponding to the right edge **31** of the first display module **30A** will be given, however, the front cover **70** may extend up to outer locations than the four edges E of the substrate **40** in the second direction Y or the third direction Z.

(161) The side end **75** of the front cover **70**, corresponding to an edge of the front cover **70**, may extend up to the outside of the substrate **40** farther than the four edges E of the substrate **40**, that is, up to the outside of the mounting surface **41**, in the second direction Y or the third direction Z.

(162) The front cover **70** may include a plurality of layers having different optical properties. The plurality of layers may have a structure resulting from stacking layers in the first direction X.

(163) The plurality of layers may constitute the front cover **70** by being bonded in the first direction X.

(164) In an embodiment, one of the plurality of layers may be an anti-glare layer, for example. As other examples, one of the plurality of layers may be an anti-reflective layer or a layer formed by combining an anti-glare layer with an anti-reflective layer.

(165) In an embodiment, another one of the plurality of layers may be a light transmittance adjustable layer. In an example embodiment, the layer may be a layer having another physical property or including another material, or a layer having another function. For example, the layer may be a circularly polarized layer. Other layers or combinations of layers may also be part of the plurality of layers

(166) In an embodiment, the front cover **70** may include a single layer, instead of the plurality of layers. The single layer may be a layer capable of implementing all functions of the plurality of layers, similar functions or some subset of functions of the plurality of layers.

(167) As described above, the front cover **70** may include an adhesive layer. The adhesive layer may be positioned at a hindmost location of the plurality of layers in the first direction X and bonded on the mounting surface **41**. The adhesive layer may have a height that is greater than or equal to a preset height in the first direction X in which the mounting surface **41** or the light-emitting surface **54** faces. Having an adhesive layer of this height may allow the adhesive layer bonded on the substrate **40** to sufficiently fill a gap that may be formed between the adhesive layer

and the plurality of inorganic light-emitting diodes **50**.

(168) However, the adhesive layer is not limited to this embodiment. For example, in another embodiment, the adhesive layer may be positioned as a separate component from the front cover **70** between the front cover **70** and the mounting surface **41** to bond the front cover **70** with the mounting surface **41**.

(169) Accordingly, because the front cover **70** is bonded with the mounting surface **41** in such a way as to be in close contact with the mounting surface **41** and protects components mounted on the mounting surface **41**, the front cover **70** may be bonded directly to the substrate **40** without any additional molding component formed between the front cover **70** and the substrate **40**.

(170) The front cover **70** may diffuse and reflect light received from the outside to prevent the light from being specularly reflected, which may be displeasing to a user.

(171) By diffusing and reflecting light received from the outside, a glaring phenomenon may be reduced, and accordingly, contrast of a screen displayed on the display panel **20** may be improved.

(172) Also, the front cover **70** may reduce transmittance of incident external light or external light reflected from the substrate **40** and the gap G.

(173) The front cover **70** according to an embodiment may include a material capable of reducing transmittance of light, to absorb at least one part of light transmitted toward the substrate **40** or light reflected from the substrate **40** and then traveling toward the first direction X.

(174) Some of a plurality of substrates may be manufactured with different colors due to processing. Accordingly, substrates having different unique colors may be tiled to constitute a single display panel.

(175) As described above, the front cover **70** according to an embodiment may absorb at least one part of light reflected from the substrate **40** and transmitted to the outside, thereby raising a sense of unity of a screen displayed on the display panel **20**.

(176) That is, according to embodiments, the front cover **70** may reduce color deviation of the display modules **30A** to **30P** generated during processes of the plurality of display modules **30A** to **30P** by lowering transmittance with respect to external light.

(177) The front cover **70** may prevent external light that enters the display panel **20** from the outside from being transmitted to the substrate **40**, and additionally absorb a part of light entered the display panel **20** from the outside or a part of external light reflected from the substrate **40** and then transmitted to the outside of the display panel **20**, thereby improving contrast of a screen that is displayed on the display panel **20**. Such different optical actions may be respectively implemented by the plurality of layers described above.

(178) That is, the front cover **70** may be positioned in front of the substrate **40** in the first direction X to improve contrast that may deteriorate by external light in a screen displayed on the display panel **20**.

(179) As described above, in the display module **30** according to an embodiment, the front cover **70** may extend up to the outside of the substrate **40** in the second direction Y.

(180) Accordingly, a part of light entered the gap G formed between the plurality of display modules **30A** to **30P** may be blocked by at least one portion of the front cover **70** positioned in the gap G, and at least a part of external light entered the gap G or reflected in the gap G may be absorbed by the front cover **70** positioned in the gap G and thus will not be transmitted to the outside. Accordingly, visibility of a seam that is formed in the gap G may be lessened, and due to the lessening visibility of the seam, there may be an improvement in a sense unity of a screen that is displayed on the display panel **20**.

(181) More specifically, the side end **75** of the front cover **70** in the second direction Y may be positioned at a more outward location than the edge **41S** of the mounting surface **41** in the second direction Y, or above the gap G.

(182) Accordingly, the front cover **70** may include a first area **71** positioned at a location more outward than the edge **41S** of the mounting surface **41** in the second direction Y or above the gap

G, and a second area **72** positioned above the mounting surface **41**.

(183) The first area **71** and the second area **72** of the front cover **70** may be partitioned by the gap G in the second direction Y.

(184) Because the first area **71** of the front cover **70** is positioned above the gap G, external light irradiated toward the gap G may be blocked by the first area **71** of the front cover **70** or light reflected in the gap G may be prevented from being irradiated to the outside. Accordingly, visibility of a seam which is a boundary between the plurality of display modules **30A** to **30P** and which may be formed by the gap G may be reduced, resulting in an improvement of a sense of unity of the display panel **20**.

(185) Because the front cover **70** extends to more outward locations than the four edges **41S** of the mounting surface **41** of the substrate **40**, as described above, visibility of seams that may be formed at the edges of the plurality of display modules **30A** to **30P** may be reduced.

(186) In an example embodiment of the first display module **30A** and the second display module **30E**, a first area **71A** of the first front cover **70A** extending from the first display module **30A** may be positioned in a gap G formed between the first display module **30A** and the second display module **30E**.

(187) Above the gap G, neighboring side ends **75A** and **75E** of the first and second front covers **70A** and **70E** of the first and second display modules **30A** and **30E** may be positioned.

(188) Also, in the gap G, the side surfaces **45** and chamfer portions **49** of the first and second display modules **30A** and **30E** may be positioned.

(189) A second area **72A** of the first front cover **70A** may be positioned above the mounting surface **41** of the first display module **30A**.

(190) A first area **71E** of the second front cover **70E**, extending from the second display module **30E**, may be positioned above the gap G formed between the first display module **30A** and the second display module **30E**, and a second area **72E** of the second front cover **70E** may be positioned above the mounting surface **41** of the second display module **30E**.

(191) That is, in the gap G formed between the first display module **30A** and the second display module **30E**, the first areas **71A** and **71E** of the first and second front covers **70A** and **70E** may be positioned side by side in the second direction Y.

(192) A length in the third direction Y of the first area **71A** or **71E** of the first or second front cover **70A** or **70B** may be substantially smaller than or equal to half of a length of the gap G.

(193) Accordingly, a sum of the lengths of the first areas **71A** and **71E** of the first and second front covers **70A** and **70E**, arranged side by side in the second direction Y, may substantially correspond to or be smaller than the length of the gap G.

(194) According to an embodiment, in the case in which the first areas **71A** and **71E** of the first and second front covers **70A** and **70E** are arranged side by side in the second direction Y, the side end **75A** of the first front cover **70A** may be spaced from the side end **75E** of the second front cover **70E**. Such spacing may be caused by the side end member **100** (which will be described below) positioned at the side ends of the display modules **30A** to **30P**.

(195) In an embodiment, the spacing may be a very small value such that it is negligible. Accordingly, in the embodiment, the first display module **30A** and the second display module **30E** may be tiled without any substantial space between the first area **71A** of the first front cover **70A** and the first area **71E** of the second cover **70E**.

(196) The first area **71A** of the first front cover **70A** and the first area **71E** of the second front cover **70E** may be positioned above the gap G between the first display module **30A** and the second display module **30E**, as described above.

(197) External light that enters the display panel **20** may be diffused and reflected to the outside of the display panel **20** while being transmitted through the first areas **71A** and **71E** of the first and second front covers **70A** and **70E**, or a part of the external light may be absorbed in the first areas **71A** and **71E**. Accordingly, an amount of the external light arrived at the gap G may be reduced,

which reduces visibility of a boundary between the first display module **30A** and the second display module **30E**, caused by the gap **G**.

(198) Also, light reflected in the gap **G** and then traveling to the outside of the display panel **20** may be diffused and reflected to the outside of the display panel **20** while being transmitted through the first areas **71A** and **71E** of the first and second front covers **70A** and **70E** of the first and second display modules **30A** and **30B**, or a part of the light may be absorbed in the first areas **71A** and **71E**. Accordingly, an amount of light transmitted to the outside of the display panel **20** may be reduced, which reduces visibility of the boundary between the first display module **30A** and the second display module **30E**, caused by the gap **G**.

(199) That is, by reducing an amount of external light entering the gap **G** formed between the plurality of display modules **30A** to **30P** while absorbing at least one part of the external light reflected in the gap **G**, a sense of unity of a screen displayed on the display panel **20** may be improved.

(200) Additionally, although a substrate **40A** of the first display module **30A** and a substrate **40E** of the second display module **30E** have different colors, at least one part of external light reflected from the substrates **40A** and **40E** may be absorbed in the first and second front covers **70A** and **70E**. Accordingly, unique colors of the substrates **40A** and **40E** may be not recognized from the outside, which improves a sense of unity of a screen displayed on the display panel **20**.

(201) The display module **30A** may include the side cover **90** positioned below the front cover **70** in the direction in which the mounting surface **41** faces and provided on the side surface **45** of the substrate **40**.

(202) More specifically, the side cover **90** may be positioned in a space defined by a lower surface **76** of the first area **71** of the front cover **70** in the first direction **X**, a lower surface of the anisotropic conductive layer **47**, and the side surface **45** of the substrate **40** in the second direction **Y**.

(203) Also, a side end **47S** of the anisotropic conductive layer **47** of the display module **30** may be aligned with the side end **75** of the front cover **70** in the first direction **X**, which is not shown. In this case, the side cover **90** may be positioned in a space defined by the lower surface of the anisotropic conductive layer **47** in the first direction **X** and the side surface **45** of the substrate **40** in the second direction **Y**.

(204) Also, the side end **47S** of the anisotropic conductive layer **47** of the display module **30** may be aligned with the edge **41S** of the mounting surface **41** in the first direction **X**. In an embodiment, the side cover **90** may be positioned in a space defined by the lower surface **76** of the first area **71** of the front cover **70** in the first direction **X** and the side surface **45** of the substrate **40** in the second direction **Y**.

(205) The side cover **90** may be bonded with the lower surface **76** of the first area **71**, the side surface **45**, and at least one portion of the side wiring **46**. The side cover **90** may be bonded with the entire of the lower surface **76** of the first area **71**. Also, the side cover **90** may cover the entire area of the side surface **45**.

(206) The lower surface **76** of the first area **71**, which may be at least one portion of the lower surface of the front cover **70**, may be a rear surface of an adhesive layer formed as the hindmost layer of the front cover **70**.

(207) In an embodiment, the side cover **90** may cover all of a pair of chamfer portions **49** positioned in a front-rear direction of the side surface **45**, in the first direction **X**.

(208) In an embodiment, the side cover **90** may surround the entire of the chamber portion **49** formed between the mounting surface **41** and the side surface **45**, as well as the side surface **45**.

(209) Because the side cover **90** of an embodiment surrounds the chamfer portion **49** formed between the mounting surface **41** and the side surface **45**, the side cover **90** may fill the entire of a space that may be formed between the substrate **40** and the front cover **70**.

(210) Accordingly, the side cover **90** of the embodiment may prevent a foreign material or water

from the outside from entering the space between the substrate **40** and the front cover **70**.

(211) Also, because the side cover **90** surrounds the chamfer portion **49** formed between the rear surface **43** and the side surface, the side cover **90** may fill the entire of a space that may be formed between the substrate **40** and the metal plate **60**.

(212) Accordingly, the side cover **90** may prevent a foreign material or water from the outside from entering the space between the substrate **40** and the metal plate **60**.

(213) The side cover **90** may be in contact with the lower surface **76** of the first area **71**, the chamfer portion **49** of the substrate **40**, and the side surface **45**. Accordingly, the side cover **90** may support the lower surface **76** of the first area **71**, the chamfer portion **49** of the substrate **40**, and the side surface **45**.

(214) The substrate **40** may be bonded with the front cover **70** as described above, and the side cover **90** may reinforce adhesiveness between the front cover **70** and the substrate **40**. Accordingly, the side cover **90** may prevent the front cover **70** from escaping from the substrate **40**.

(215) That is, the side cover **90** may increase reliability of the display module **30A**.

(216) Also, the substrate **40** may be bonded with the metal plate **60** by the rear adhesive tape **61**, and the side cover **90** may reinforce adhesiveness between the substrate **40** and the metal plate **60**. Accordingly, the side cover **90** may prevent the metal plate **60** from escaping from the substrate **40**.

(217) The side surface **45** of the substrate **40** may correspond to the four edges **41S** of the mounting surface **41**, as described above, and the first area **71** of the front cover **70** may extend to locations more outward than the four edges **41S** of the mounting surface **41** in the second direction Y and the third direction Z in which the mounting surface **41** extends.

(218) The side cover **90** may surround the lower surface **76** of the first area **71** and the side surface **45** corresponding to each of the four edges **41S** of the mounting surface **41**, along the four edges **41S** of the mounting surface **41**.

(219) That is, the side cover **90** may seal all edges of a portion at which the substrate **40** is bonded with the front cover **70**.

(220) The side cover **45** may cover the lower surface **76** of the first area **71** and the side surface **45** in all directions that are orthogonal to the first direction X.

(221) Accordingly, bonding between the front cover **70** and the substrate **40** may be improved, and the front cover **70** and the side surface **45** of the substrate **40** may be protected from an external force.

(222) Also, outside water or a foreign material may be prevented from entering between the substrate **40** and the front cover **70**, as described above. In addition, upon formation of a gap between the substrate **40** and the front cover **70** due to degradation of adhesion, outside water or a foreign material may be prevented from entering the gap.

(223) The side cover **90** may surround all four edges E of the substrate **40** along the side surface **45** of the substrate **40** to seal between the substrate **40**, the front cover **70**, and the metal plate **60**.

(224) Accordingly, the side cover **90** may prevent a foreign material or water entered the substrate **40** in all directions from permeating into the substrate **40** and the front cover **70**.

(225) Because the hindmost layer of the front cover **70** in the first direction X is an adhesive layer, as described above, the lower surface **76** of the first area **71** may be a rear surface of the adhesive layer.

(226) Accordingly, upon exposure of the lower surface **76** of the first area **71** to the outside, a foreign material floating in the outside may be adhered to the lower surface **76** of the first area **71**.

(227) Upon arraying of the plurality of display modules **30A** to **30P** in a state in which a foreign material is adhered to the lower surface **76** of the first area **71**, visibility of a seam generated between the plurality of display modules **30A** to **30P** may be increased by the foreign material adhered to the lower surface **76** of the first area **71**.

(228) However, because the display module **30A** according to an embodiment includes the side cover **90** and the side cover **90** covers the lower surface **76** of the first area **71**, the display module

30A may prevent a foreign material from being adhered to the lower surface **76** of the first area **71**.
(229) Accordingly, visibility of a seam generated between the plurality of display modules **30A** to **30P** due to a foreign material adhered to the front cover **70** upon arraying of the plurality of display modules **30A** to **30P** may be reduced.

(230) Also, current may flow to a plurality of electronic components mounted on the substrate **40** by an electrostatic discharge that occurs on the display modules **30A** to **30P** to damage the electronic components. The side cover **90** may seal the substrate **40** from the outside and thus block charges generated by an electrostatic discharge from entering the substrate **40**, to prevent the electronic components from being damaged, which will be described later.

(231) That is, because the substrate **40** is sealed by the front cover **70** and the side cover **90**, charges generated by an electrostatic discharge may be prevented from passing through the front cover **70** and the side cover **90** and thus flowing to the substrate **40**. Also, charges flowing on the front cover **70** and the side cover **90** may be guided to the metal plate **60** being in contact with the side cover **90**, thereby providing a path for current generated by an electrostatic discharge. Accordingly, electrostatic discharge (ESD) internal pressure of the electronic components mounted on the substrate **40** may be improved.

(232) As described above, the display module **30A** may be positioned below the front cover **70** in the direction in which the mounting surface **41** faces. That is, the side cover **90** may be not positioned above the lower surface **76** in the first direction X.

(233) A foremost surface of the side cover **90** in the first direction X may be in contact with the lower surface **76** of the first area **71**, and may be not positioned before the lower surface **76** of the first area **71** in the first direction X. According to an embodiment, this may be done so that the side cover **90** is not on a traveling path of light emitted from the plurality of inorganic light-emitting diodes **50**.

(234) In a case in which at least one portion of the side cover **90** is positioned before the lower surface **76** or the front cover **70** in the first direction X, the at least one portion of the side cover **90** may be positioned on a traveling path of light traveling forward through the front cover **70**.

(235) That is, the side cover **90** may absorb or diffuse and reflect a part of traveling light to distort an area of an image displayed on the display panel **20**.

(236) However, because the side cover **90** according to an embodiment is positioned behind the front cover **70** in the first direction X, the side cover **90** may not limit the travel of light emitted from the plurality of light-emitting diodes **50**, thereby improving image quality of the display panel **20**.

(237) The side end **75** of the front cover **70** in the second direction Y and a side end portion **91** of the side cover **90** in the second direction Y may be substantially aligned in the first direction X.

(238) The reason may be because the front cover **70** and the side cover **90** are cut simultaneously in a process of manufacturing the display module **30A**. Also, the side end member **100** may be bonded with the side end **75** of the front cover **70** and the side end portion **91** of the side cover **90**, substantially aligned in the first direction X.

(239) That is, a space that may be formed between the plurality of display modules **30A** to **30P** upon arraying of the plurality of display modules **30A** to **30P** may be reduced, and visibility of a seam that is caused by the space between the plurality of display modules **30A** to **30P** may be reduced.

(240) The side cover **90** may include a light absorbing material. For example, the side cover **90** may be formed of an opaque or translucent material.

(241) Also, the side cover **90** may include a photosensitive material. For example, the side cover **90** may be formed of a photosensitive OCR. The photosensitive material may change physical properties to show a dark color by receiving external light, such as ultraviolet (UV) light, having a wavelength that is different from a wavelength of visible light.

(242) Accordingly, by irradiating UV light to the side cover **90** during a manufacturing process to

color the side cover **90** with a dark color, the side cover **90** may be provided as a light absorbing member.

(243) The side cover **90** may have a dark color. The side cover **90** may have a darker color than the front cover **70**.

(244) The side cover **90** may have a similar color to that of the black matrix **48**.

(245) In the embodiment, owing at least in part to the color of the side cover **90**, at least a portion of light may be absorbed in the side cover **90** without being reflected, by the light absorbing member of the side cover **90**.

(246) The side cover **90** may be positioned in the gap G formed between the plurality of display modules **30A** to **30P**, together with the first area **71** of the front cover **70**, upon arraying of the plurality of display modules **30A** to **30P**.

(247) Accordingly, by absorbing light entered the gap G, an amount of light reflected and then emitted to the outside among the light entered the gap G may be reduced. Therefore, visibility of a seam formed by the gap G between the plurality of display modules **30A** to **30P** may be reduced.

(248) In the example of the first display module **30A** and the second display module **30E**, the first side cover **90A** of the first display module **30A** and a second side cover **90E** of the second display module **30E** may be positioned in the gap G formed between the first display module **30A** and the second display module **30E**, together with the first area **71A** of the first front cover **70A** and the first area **71E** of the second front cover **70E**.

(249) In the gap G, the side end portion **91A** of the side cover **90A** of the first display module **30A** and the side end portion **91E** of the side cover **90E** of the second display module **30E** may be positioned together with the neighboring side ends **75A** and **75E** of the first and second front covers **70A** and **70E** of the first and second display modules **30A** and **30E**.

(250) The neighboring side ends **75A** and **75E** of the first and second front covers **70A** and **70E** of the first and second display modules **30A** and **30E** and the neighboring side end portions **91A** and **91E** of the side covers **90A** and **90E** of the first and second display modules **30A** and **30E** may be opposite to each other. The neighboring side ends **75A** and **75E** of the first and second front covers **70A** and **70E** of the first and second display modules **30A** and **30E** and the neighboring side end portions **91A** and **91E** of the side covers **90A** and **90E** of the first and second display modules **30A** and **30E** may be positioned in parallel to each other.

(251) That is, in the gap G formed between the first display module **30A** and the second display module **30E**, the first areas **71A** and **71E** of the first and second front covers **70A** and **70E** and the side covers **90A** and **90E** of the first and second display modules **30A** and **30E** may be positioned side by side in the second direction Y.

(252) A length in second direction Y of each of the side covers **90A** and **90E** of the first and second display modules **30A** and **30E** may be substantially smaller than or equal to half of the length of the gap G to correspond to each of the first areas **71A** and **71E** of the first and second front covers **70A** and **70E**.

(253) In the gap G between the first display module **30A** and the second display module **30E**, the first area **71A** of the first front cover **70A** and the first area **71E** of the second front cover **70E** may be positioned, and the side covers **90A** and **90E** of the first and second display modules **30A** and **30E** may be positioned behind the first areas **71A** and **71E** in the first direction X.

(254) External light entered the display panel **20** may be diffused and reflected to the outside of the display panel **20** by being transmitted through the first areas **71A** and **71E** of the first and second front covers **70A** and **70E**, or a part of the external light may be absorbed in the first areas **71A** and **71E** of the first and second front covers **70A** and **70E**. Accordingly, an amount of light arrived at the gap G may be reduced.

(255) In addition, light arrived at the gap G may be absorbed in the side covers **90A** and **90E** of the first and second display modules **30A** and **30E**, positioned in the gap G, and accordingly, visibility of a boundary between the first display module **30A** and the second display module **30E** may be

reduced.

(256) That is, by reducing an amount of external light entering the gap G formed between the plurality of display modules **30A** to **30P** and additionally absorbing light arrived at the gap G, a sense of unity of a screen displayed on the display panel **20** may be improved.

(257) In addition, light reflected from the side covers **90A** and **90E** of the first and second display modules **30A** and **30E** and then traveling to the outside of the display panel **20** without being absorbed in the side covers **90A** and **90E** may be diffused and reflected to the outside of the display panel **20** by being transmitted through the first areas **71A** and **71E** of the first and second front covers **70A** and **70E**, or a part of the light may be absorbed in the first areas **71A** and **71E**.

Accordingly, an amount of light transmitted to the outside of the display panel **20** may be reduced, and thus, visibility of the boundary between the first display module **30A** and the second display module **30E**, caused by the gap G, may be reduced.

(258) Because the side cover **90** is positioned in the gap G formed between the plurality of display modules **30A** to **30P** upon arraying of the plurality of display modules **30A** to **30P**, as described above, the side cover **90** may absorb light arrived at the gap G to reduce visibility of a seam, caused by the gap G.

(259) In the above-described example embodiment, the front cover **70** may diffuse and reflect, absorb, or circularly polarize a part of light entered the display panel **20**, or change a reflection direction of the part of the light, thereby reducing an amount of light that arrives at the substrate **40**, although the embodiment is not limited thereto.

(260) In an embodiment, the front cover **70** may be formed of a transparent material to transmit light without any deformation. In this case, visibility of the boundary between the plurality of display modules **30A** to **30P**, caused by the gap G, may be reduced by the side cover **90** positioned between the plurality of display modules **30A** to **30P**.

(261) Because the side cover **90** is formed of a light absorbing material, as described above, a part of light emitted from the plurality of inorganic light-emitting diodes **50** may be absorbed in the side cover **90** in a case in which at least one portion of the side cover **90** is positioned before the front cover **70** in the first direction X. Accordingly, a certain area of a screen displayed on the display panel **20** may appear dark.

(262) However, because the side cover **90** according to an embodiment is positioned below the front cover **70** in the first direction X, more specifically, below the lower surface **76** of the first area **71**, the side cover **90** may not absorb light emitted from the plurality of inorganic light-emitting diodes **50**, and accordingly, an image displayed on the display panel **20** may have uniform brightness.

(263) The anisotropic conductive layer **47** may be in a form of an anisotropic conductive film. The anisotropic conductive layer **47** may be adhered in a form of a film to the TFT layer **41**.

(264) Because the anisotropic conductive layer **47** is in the form of the film, an area of the anisotropic conductive layer **47** may be larger than an area of the substrate **40**.

(265) Accordingly, after the anisotropic conductive layer **47** is adhered to the TFT layer **44**, a process of cutting the anisotropic conductive layer **47** to correspond the area of the anisotropic conductive layer **47** to the area of the substrate **40** may be performed.

(266) The cutting process may be to cut the anisotropic conductive layer **47** through laser cutting, etc. to correspond the area of the anisotropic conductive layer **47** to the area of the substrate **40**.

(267) The anisotropic conductive layer **47** may have an area corresponding to an area of the mounting surface **41**. In some instances, it may be challenging to form the anisotropic conductive layer **47** as an anisotropic conductive film, as described above, of which an area corresponds to the area of the mounting surface **41**, and, upon bonding of the anisotropic conductive film having the area corresponding to the mounting surface **41** onto the mounting surface **41**, the anisotropic conductive film may have a smaller cross-sectional area than the mounting surface **41** due to manufacturing tolerance of the anisotropic conductive film, which deteriorates reliability of the

display module **30**.

(268) Accordingly, in some embodiments, an anisotropic conductive film having a larger area than the mounting surface **41** is bonded with the substrate **40** and then cutting the anisotropic conductive film to an area corresponding to the substrate **40**, the anisotropic conductive layer **47** may be formed.

(269) The side surface **45** of the substrate **40** may be positioned at an outer location than the mounting surface **41** by the chamfer portion **49**. In this embodiment, the anisotropic conductive film may be cut based on the side surface **45** of the substrate **40** in the second direction Y, although the embodiment is not limited thereto.

(270) Also, the anisotropic conductive film may be cut based on a side end of the coating member **46a** forming a side end of the side wiring **46** in the third direction Z (see FIG. **10**). The reason may be because the side surface **45** of the substrate **40**, the chamfer portion **49**, or the side wiring **46** may be damaged upon cutting of the anisotropic conductive layer based on the mounting surface **41**.

(271) However, the anisotropic conductive layer **47** may be formed by cutting the anisotropic conductive film together with the front cover **70**. The side end **47S** of the anisotropic conductive layer **47** may be aligned with the side end **75** of the front cover **70** in the first direction X.

(272) In this case, the side end **47S** of the anisotropic conductive layer **47** may be exposed to the outside, and thus may be damaged by static electricity. However, reliability against ESD may be secured by the side end member **100** which will be described below.

(273) Accordingly, upon cutting of the anisotropic conductive film, the side end **47S** of the anisotropic conductive layer **47** may be positioned at an outer location than the mounting surface **41**. More specifically, the anisotropic conductive film may be cut based on the side surface **45** or the side end of the side wiring **46**, as described above. The side end **47S** of the anisotropic conductive layer **47** may be aligned with the side surface **45** or the side end of the side wiring **46** in the first direction X. Also, due to manufacturing tolerance or burr molded in the anisotropic conductive film upon cutting, the anisotropic conductive layer **47** may be positioned at an outer location than the side surface **45** or the side end **46S** of the side wiring **46**.

(274) In an embodiment, to prevent breakage of the substrate **40**, which may actually occur in a cutting process, a location at which the anisotropic conductive film is cut may be an outer location than the side surface **45** or the side end **46S** of the side wiring **46**.

(275) Accordingly, the side end **47S** of the anisotropic conductive layer **47** may be positioned outside the substrate **40**. Particularly, the side end **47S** of the anisotropic conductive layer **47** may be positioned at the outer location than the side cover **90**.

(276) The side cover **90** may cover both an outer side in third direction Z of the side surface **45** of the substrate **40** and an outer side in second direction Y of the side surface **45**, as shown in FIGS. **7** to **10**.

(277) That is, the side cover **90** may surround all of the four edges E of the substrate **40**, as described above.

(278) Accordingly, the mounting surface **41** which is the front surface of the substrate **40** may be covered by the front cover **70**, the rear surface **43** of the substrate **40** may be covered by the metal plate **60**, and the side surface **45** and the chamfer portion **49** of the substrate **40** may be covered by the side cover **90**.

(279) The front cover **70** may be formed of a nonconductive material through which no charges are transmitted.

(280) In an embodiment, the side cover **90** may be formed of a nonconductive material through which no charges are transmitted.

(281) In an embodiment, because the front cover **70** and the side cover **90** are formed of a nonconductive material, a major portion of current applied to the front cover **70** or the side cover **90** may float on the front cover **70** and the side cover **90** without being transmitted through the

front cover **70** and the side cover **90**.

(282) Also, the metal plate **60** may be formed of a material having a high capacitance, and function as a ground component. Accordingly, upon application of current to the metal plate **60**, the metal plate **60** may be maintained at a constant potential, and the current applied to the metal plate **60** may be absorbed in the metal plate **60** without flowing to the substrate **40** through the metal plate **60**.

(283) Also, the entire of the side wiring **46** of the substrate **40** may be surrounded by the side cover **90**, and accordingly, the side wiring **46** may be sealed without being exposed to the outside.

Therefore, although an electrostatic discharge occurs at the side surface **45** of the substrate **40**, no current may enter the side wiring **46** by the side cover **90**.

(284) In a process of manufacturing a display apparatus by implementing a display panel with display modules, a plurality of display modules may be tiled to form the display panel. In a process of forming the display panel with the display modules, current generated by an electrostatic discharge while the display modules are manufactured and delivered may enter the insides of the display modules to damage electronic components installed inside the display modules.

(285) Particularly, during a process of manufacturing the display module **30**, a defect may be generated, and in this case, the side wiring **46** extending along the side surface **45** of the substrate **40** may be exposed to the outside, or a space may be made between the anisotropic conductive layer **47** and the front cover **70** or the substrate **40**, and an inside space may be made in a process of applying and hardening the side cover **90**. In this case, according to an electrostatic discharge, current may enter electronic components, such as the side wiring **46**, etc., mounted on the substrate **40**, due to the defect, to damage the electronic components.

(286) To prevent current generated by an electrostatic discharge from entering the display module **30** to damage electronic components installed inside the display module **30** in a process before the display module **30** is coupled with the frame **15** to be assembled into the display apparatus **1**, the display module **30** may include the front cover **70**, the side cover **90**, and the metal plate **60** to absorb an electrical impact.

(287) Accordingly, each of the display modules **30A** to **30P** may independently include a component for blocking current generated by an electrostatic discharge from entering components mounted on the substrate **40**, and the current generated by the electrostatic discharge may be easily guided to the metal plate **60** which is a ground component along the front cover **70** and the side cover **90** sealing the substrate **40** on each of the display modules **30A** to **30P** without entering the components mounted on the substrate **40**.

(288) The display apparatus **1** according to an embodiment may further include the side end member **100** positioned on an outer side end of the side cover **90** in the second direction **Y** and the third direction **Z** of the display module **30** and formed of a material having higher conductivity than the side cover **90**.

(289) The side end member **100** may easily guide static electricity to the metal plate **60** although the display modules **30A** to **30P** are insufficiently sealed due to a defect caused during a manufacturing process.

(290) The side end member **100** may cover the outer side in second direction **Y** and third direction **Z** of the side surface **45** of the substrate **40**, as shown in FIGS. **7** to **10**. That is, the side end member **100** may surround all of the four edges of the substrate **40**, although not limited thereto.

(291) However, the side end member **100** may be formed only on the side surface **45** on which the side wiring **46** is formed. That is, according to an embodiment, the side end member **100** may be positioned only on the side surfaces **45** corresponding to the upper edge **32** and the lower edge **34** on which the side wiring **46** is formed.

(292) The side end member **100** may be formed of a metal material having higher conductivity than the side cover **90**. The side end member **100** may be coated on the side cover **90** and positioned on the outer side end of the side cover **90**.

(293) Accordingly, the side end member **100** may be positioned in the gap G formed between the display modules **30A** to **30P** upon arraying of the display modules **30A** to **30P**.

(294) One end of the side end member **100** may be in contact with the metal plate **60**, and the other end of the side end member **100** may be positioned on the side end portion **91** of the side cover **90**. That is, the side end member **100** may cover at least one of the side cover **90** and at least one portion of the metal plate **60** in the second direction Y and the third direction Z.

(295) The side end member **100** may be in a form of a thin film. The reason may be because the side end member **100** is positioned in the gap G formed between the display modules **30A** to **30P** upon tiling of the display modules **30A** to **30P**. A great thickness of the side end member **100** may require a great gap G between the display modules **30A** to **30P**, which may cause visibility of a seam between the display modules **30A** to **30P**.

(296) A certain space may be generated between the neighboring display modules **30A** to **30P** by the side end member **100** although the side end member **100** is in the form of a thin film. To reduce the certain space between the neighboring display modules **30A** to **30P**, neighboring side end members **100** may overlap with each other in the second direction Y and the third direction Z, which will be described in detail below.

(297) The side end member **100** may be formed of a material having high conductivity. For example, the side end member **100** may be formed of a material, such as a metal, a conductive polymer, a conductive fabric, etc., which is capable of being electrically grounded to the metal plate **60**.

(298) The side end member **100** may be formed of a material having higher conductivity than the side cover **90**. Also, the side end member **100** may be formed of a material having higher conductivity than the front cover **70**.

(299) Accordingly, current generated by an electrostatic discharge E1 on the front cover **70** or the side cover **90** may not enter the substrate **40** because the current is not transmitted through the front cover **70** or the side cover **90**, and the current may float on the front cover **70** to enter the side end member **100**.

(300) The current entered the side end member **100** may enter the metal plate **60** through the side end member **100**. The reason may be because the side end member **100** is in contact with the metal plate **60** and grounded to the ground component.

(301) That is, the side end member **100** may provide a current path for causing current generated by an electrostatic discharge occurred on the front cover **70** or the side cover **90** to flow to the metal plate **60** provided as the ground component. The side end member **100** may guide charges generated by an electrostatic discharge to the ground.

(302) Accordingly, because a major part of current generated by an electrostatic discharge on the front cover **70** or the side cover **90** flows to the metal plate **60** through the side end member **100** having high conductivity, ESD internal pressure of the electronic components mounted on the substrate **40** may be improved, although some current flows to the substrate **40**.

(303) In addition, electrostatic current transferred to the metal plate **60** may escape to an external ground through a component, such as a bridge board or a cable, being in contact with the metal plate **60**.

(304) The side end member **100** may have a dark color. The side end member **100** may have a black color. The side end member **100** may have a darker color than the front cover **70**.

(305) The side end member **100** may have a similar color to that of the black matrix **48** or the side cover **90**. Accordingly, light entered the side end member **100** may be absorbed in the side end member **100** without being reflected.

(306) As described above, each of the display modules **30A** to **30P** may independently include the front cover **70**, the side cover **90**, the metal plate **60**, and the side end member **100** to prevent infiltration of current generated according to an electrostatic discharge.

(307) Accordingly, after the front cover **70**, the side cover **90**, the metal plate **60**, and the side end

member **100** are installed in each of the display modules **30A** to **30P**, the display modules **30A** to **30P** may be protected from current generated according to an electrostatic discharge that may occur in the subsequent manufacturing process or a transfer process. Particularly, current generated by an electrostatic discharge may be easily guided to the metal plate **60** by the side end member **100** although the front cover **70** and the side cover **90** fail to completely seal the substrate **40** due to a defect caused during a manufacturing process, and accordingly, the display modules **30A** to **30P** may be protected from current generated by an electrostatic discharge.

(308) Hereinafter, a shape of the side end member **100**, capable of reducing a space that is generated between the neighboring display modules **30A** to **30P** by the side end member **100**, will be described in detail. For convenience of description, the side end member **100** will be described with reference to the first display module **30A**, though each of the display modules **30A** to **30P** of the example embodiment include a side end member **100** and/or side end member **100'**.

(309) As shown in FIG. **8**, the side end member **100** may include a first end portion **120** and a second end portion **130** spaced from the first end portion **120** in the first direction **X**.

(310) More specifically, the first end portion **120** may be positioned on the side end portion **91** of the side cover **90** such that the side end member **100** covers at least one portion of the side cover **90**. The second end portion **130** may cover at least one portion of the metal plate **60** and be in contact with the metal plate **60** such that the side end member **100** is grounded to the metal plate **60**. As shown, the first end portion **120** may be positioned before the second end portion **130** in the first direction **X**. That is, for example, the first end portion **120** may be relatively closer to the front cover **170** and the second end portion **130** relatively farther from the cover in the first direction **X**.

(311) The first end portion **120** may be positioned behind the front cover **70** in the first direction **X**. The first end portion **120** may be positioned behind the light-emitting surface **54** of the plurality of inorganic light-emitting diodes **50** in the first direction **X**.

(312) The reason may be because the first end portion **120** of the side end member **100** positioned in the gap **G** above the light-emitting surface **54** may be recognized as a seam that may be generated between the display modules **30A** to **30P** by light irradiated from the plurality of inorganic light-emitting diodes **50**.

(313) The side end member **100** may include a side end portion **110** extending from the first end portion **120** to the second end portion **130** and forming a side end of the side end member **100**.

(314) One surface of the side end member **100** may be in contact with the side cover **90** and the metal plate **60**, and the other surface of the side end member **100** may be positioned outward in an extension direction of the mounting surface **41** and form the side end portion **110**.

(315) The side end portion **110** may include a plurality of mountains **111** and a plurality of valleys **112** that are alternately arranged in the first direction **X**.

(316) The plurality of mountains **111** may be protrusions protruding in the second direction **Y** or the third direction **Z**, and the plurality of valleys **112** may be concave in the second direction **Y** or the third direction **Z** with respect to the plurality of mountains **111**.

(317) Because the plurality of mountains **111** and the plurality of valleys **112** are alternately arranged, the plurality of valleys **112** may form concave grooves **113**.

(318) A mountain **121** among the plurality of mountains **111** and the plurality of valleys **112** may be positioned at the first end portion **120**. The mountain **121** positioned in the first end portion **120** is referred to as a first mountain **121**.

(319) A valley **131** among the plurality of mountains **111** and the plurality of valleys **112** may be positioned at the second end portion **130**. The valley **131** positioned in the second end portion **130** is referred to as a first valley **131**.

(320) At a front end of the side end portion **110** in the first direction **X**, the first mountain **121** may be positioned and the plurality of mountains **111** and the plurality of valleys **112** may be alternately positioned starting from the first mountain **121**. At a rear end of the side end portion **110**, the first valley **131** may be positioned. The side end portion **110** may end at the first valley **131**.

(321) According to an embodiment, the separation distance T between the plurality of display modules **30A** to **30P** may be reduced by overlapping the side end members **100** of the display modules **30A** to **30P** with each other in the second direction Y or the third direction Z upon arraying of the display modules **30A** to **30P**.

(322) For example, in the side end member **100** of the first display module **30A**, the first end portion **120** may be positioned at the front end of the side end member **100** in the first direction X and the second end portion **130** may be positioned at the rear end of the side end member **100**, as described above.

(323) As shown in FIG. 7, the side end member **100** of the second display module **30E** being adjacent to the first display module **30A** in the second direction Y may be positioned in a state of being rotated to 180 degrees in the first direction X with respect to the side end member **100** of the first display module **30A**. The side end members **100** of all display modules that are adjacent to the first display module **30A** in the second direction Y and the third direction Z, such as the second display module **30E** shown in FIG. 7 and the third display module **30B** shown in FIG. 10, may be positioned in a state of being rotated to 180 degrees in the first direction X with respect to the side end member **100** of the first display module **30A**.

(324) That is, in the side end member **100** of the second display module **30E**, the first end portion **120** may be positioned at a rear end of the side end member **100** in the first direction X, and the second end portion **130** may be positioned at a front end of the side end member **100** in the first direction X.

(325) In the side end member **100** of the first display module **30A**, the first end portion **120** is positioned at the front end of the side end member **100** in the first direction X and the second end portion **130** is positioned at the rear end of the side end member **100**. Other display modules of the plurality of display modules **30A** to **30P** may include side member **100'**. Side end member **100'** should be assumed to include the same features as side end member **100** unless otherwise described and the description of features of side end member **100** are applicable to side end member **100'**. In side end member **100'**, the first end portion **120** is positioned at the rear end of the side end member **100** in the first direction X and the second end portion **130** is positioned at the front end of the side end member **100** in the first direction X is defined as a side end member **100'** of a second arrangement. The second display module **30E** includes side end member **100'** as shown in FIG. 7. As shown in the embodiment of FIG. 7, because the first display module **30A** has the side end member **100** and the second display module **30E** has the side end member **100'**, the first display module **30A** and the second display module **30E** may be engaged at the side end members **100** and **100'**. In particular, the mountains **111** of the side end member **100** are positioned to be inserted into the valleys **112** of the side end member **100'**. Conversely, the mountings **111** of the side end member **100'** inserted into the valleys **112** of the side end member **100** thereby engaging first display module **30A** and second display module **30E** through the respective side members **100**, **100'**.

(326) In embodiments, any of the plurality of display modules **30A** to **30P** may have either or both of the side end members **100** and **100'**. For example, in another example embodiment, instead of the first display module **30A** including a side end module **100** and the second display module **30E** including a side end module **100'**, they may be reversed such that the first display module **30A** includes a side end module **100'** and the second display module **30E** includes a side end module **100**.

(327) Referring back to the example embodiment as shown in FIG. 7, the side end member **100'** of the second display module **30E** may be bonded with the side cover **90E** and the metal plate **60** of the second display module **30E** such that a first end portion **120'** is in contact with the metal plate **60** and a second end portion **130'** is positioned on the side end portion **91** of the side cover **90**.

(328) Because a mountain **121** is positioned in the first end portion **120'** and a valley **131** is positioned in the second end portion **130'**, as described above, the side end portion **100'** having the

second arrangement may include the valley **131** at the front end in the first direction X.

(329) Accordingly, the plurality of mountains **111** and the plurality of valleys **112** of the side end member **100** of the first display module **30A** may be arranged alternately with the plurality of mountains **111** and the plurality of valleys **112** of the side end member **100'** of the second display module **30E** at the same height in the first direction X.

(330) That is, in a case in which a mountain **111** is positioned in the side end member **100** of the first display module **30A** at a certain height in the first direction X, a valley **112** may be positioned in the side end member **100'** of the second display module **30E** at the certain height in the first direction X. Inversely, in a case in which a valley **112** is positioned in the side end member **100** of the first display module **30A** at a certain height in the first direction X, a mountain **111** may be positioned in the side end member **100'** of the second display module **30E** at the certain height in the first direction X.

(331) Accordingly, upon arraying of the first display module **30A** and the second display module **30E** in the second direction Y, the side end member **100** of the first display module **30A** may overlap with the side end member **100'** of the second display module **30E** in the second direction Y.

(332) That is, the plurality of mountains **111** of the side end member **100** of the first display module **30A** may be inserted into the grooves **113** formed by the plurality of valleys **112** of the side end member **100'** of the second display module **30E**.

(333) Accordingly, at least one portion of the side end member **100** of the first display module **30A** may overlap with at least one portion of the side end member **100'** of the second display module **30E** in the second direction Y to form a coupling section C, and a separation distance T between the first display module **30A** and the second display module **30E** may be reduced by a length of the overlapping portion of the coupling section C in the second direction Y.

(334) A protrusion height of the plurality of mountains **111** of the side end member **100** may be equal to a depth of the plurality of valleys **112**. The may be done so that the plurality of mountains **111** of the first display module **30A** can be inserted into the corresponding plurality of valleys **112** of the second display module **30E**. Similarly, the plurality of mountains **111** of the second display module **30E** can be inserted into the corresponding plurality of valleys **112** of the first display module **30A** and a constant maximum depth upon overlapping of the at least one portion of the side end member **100** of the first display module **30A** with the at least one portion of the side end member **100'** of the second display module **30E** is maintained.

(335) Also, a protrusion volume of each of the plurality of mountains **111** may be equal to a depression volume of each of the plurality of valleys **112**. The may be done so that the plurality of mountains **111** of the first display module **30A** can be inserted into the corresponding plurality of valleys of the second display module **30E**. Similarly, the plurality of mountains **111** of the second display module **30E** can be inserted into the corresponding plurality of valleys **112** of the first display module **30A** while maintaining a constant maximum volume.

(336) In the side end member **100**, a base length of the side end member **100** is defined as t1, and a protrusion length of each of the plurality of mountains **111** in the second direction Y or the third direction Z is defined as t2. In this case, a separation distance T between the first display module **30A** and the second display module **30E** may be $2*t1+t2$. In a case in which the side end portion **110** of the side end member **100** is not formed as the plurality of mountains **111** and the plurality of valleys **112**, a separation distance T between the first display module **30A** and the second display module **30E** may be $2*t1+2*t2$. Therefore, due to the side end portion **110** including the plurality of mountains **111** and the plurality of valleys **112**, the separation distance T between the first display module **30A** and the second display module **30E** may be reduced by a length of t2, and accordingly, the separation distance T between the first display module **30A** and the second display module **30E** may be reduced. As shown in FIGS. **9** and **10**, the front cover **70** may extend up to an outer location than the substrate **40** in the third direction Z. More specifically, the front cover **70** may extend up to an outer location than the side surface **45** and the chamfer portion **49** in the third direction Z.

(337) The side end **75** of the front cover **70** in the third direction **Z** may be positioned at an outer location than the edge **41S** of the mounting surface **41** in the third direction **Z**, or above the gap **G**.

(338) The first area **71** and the second area **72** of the front cover **70** may also be partitioned by the gap **G** in the third direction **Z**.

(339) In the example of the first display module **70** and the third display module **30B**, the first area **71A** of the first front cover **70A** of the first display module **30A** may be positioned in a gap **G** formed between the first display module **30A** and the third display module **30B**.

(340) In the gap **G**, the neighboring side ends **75A** and **75B** of the first and third front covers **70A** and **70B** of the first and third display modules **30A** and **30B** may be positioned.

(341) Also, in the gap **G**, the side surfaces **45** and chamfer portions **49** of the first and third display modules **30A** to **30B** may be positioned.

(342) The first area **71B** of the third front cover **70B** of the third display module **30B** may be positioned in the gap **G** formed between the first display module **30A** and the third display module **30B**, and a second area **72B** of the third front cover **70B** may be positioned above the mounting surface **41** of the third display module **30B**.

(343) That is, in the gap **G** formed between the first display module **30** and the third display module **30B**, the first areas **71A** and **71B** of the first and third front covers **70A** and **70B** may be positioned side by side in the third direction **Z**.

(344) External light entered the display panel **20** may be diffused and reflected to the outside of the display panel **20** while being transmitted through the first areas **71A** and **71B** of the first and third front covers **70A** and **70B**, or a part of the external light may be absorbed in the first areas **71A** and **71B**. Accordingly, an amount of the external light arrived at the gap **G** may be reduced, which reduces visibility of a boundary between the first display module **30A** and the third display module **30B**, caused by the gap **G**.

(345) Also, light reflected in the gap **G** and then traveling to the outside of the display panel **20** may be diffused and reflected to the outside of the display panel **20** while being transmitted through the first areas **71A** and **71B** of the first and third front covers **70A** and **70B**, or a part of the light may be absorbed in the first areas **71A** and **71B**. Accordingly, an amount of light transmitted to the outside of the display panel **20** may be reduced, which reduces visibility of the boundary between the first display module **30A** and the third display module **30B**, caused by the gap **G**.

(346) As described above, the side cover **90** may be positioned in a space formed on the side surface **45** of the substrate **40** in the third direction **Z**, as well as the second direction **Y**.

(347) On the side surface **45** of the substrate **40**, positioned toward the third direction **Z**, the side wiring **46** may be positioned. Accordingly, the side cover **90** provided on the side surface **45** positioned toward the third direction **Z** may surround the side wiring **46**, as well as the side surface **45** and the chamfer portion **49**. Accordingly, the side cover **90** may protect the side wiring **46** from an external force and prevent a foreign material or water from permeating into the side wiring **46**.

(348) That is, the side cover **90** may surround the lower surface **76** of the first area **71** and the side surface **45** corresponding to each of the four edges **41S** of the mounting surface **41**, along the four edges **41S** of the mounting surface **41**, and surround the side wiring **46** extending along the side surface **45** in the third direction **Z**.

(349) Accordingly, bonding between the front cover **70** and the substrate **40** may be improved, and the front cover **70**, the side surface **45** of the substrate **40**, and the side wiring **46** may be protected from an external force.

(350) The side end **75** in third direction **Z** of the front cover **70** may be aligned with the side end portion **91** in third direction **Z** of the side cover **90** in the first direction **X**. The side end **75** in third direction **Z** of the front cover **70** may be aligned with the side end portion **91** in third direction **Z** of the side cover **90** in a direction being parallel to the first direction **X**.

(351) In the example of the first display module **30A** and the second display module **30E**, the first side cover **90A** of the first display module **30A** and the third side cover **90B** of the third display

module **30B** may be positioned in the gap **G** formed between the first display module **30A** and the third display module **30B**, together with the first area **71A** of the first front cover **70A** and the first area **71B** of the third front cover **70B**.

(352) In the gap **G**, the neighboring side end portions **91A** and **91B** of the side covers **90A** and **90B** may be positioned together with the neighboring side ends **75A** and **75B** of the front covers **70A** and **70B** of the first and third display modules **30A** and **30B**.

(353) The neighboring side ends **75A** and **75B** of the front covers **70A** and **70B** and the neighboring side end portions **91A** and **91B** of the side covers **90A** and **90B** may be opposite to each other.

(354) The neighboring side ends **75A** and **75B** of the front covers **70A** and **70B** and the neighboring side end portions **91A** and **91B** of the side covers **90A** and **90B** may be parallel to each other.

(355) That is, in the gap **G** formed between the first display module **30A** and the third display module **30B**, the first areas **71A** and **71B** of the first and third front covers **70A** and **70B** and the first and third side covers **90A** and **90B** may be positioned side by side in the third direction **Z**.

(356) In the gap **G** formed between the first display module **30A** and the third display module **30B**, the first areas **71A** and **71B** of the first and third front covers **70A** and **70B** and the first and third side covers **90A** and **90B** may be positioned side by side in the third direction **Z**.

(357) In the gap **G** formed between the first display module **30A** and the third display module **30B**, the first area **71A** of the first front cover **70A** and the first area **71B** of the third front cover **70B** may be positioned, and the first and third side covers **90A** and **90B** may be positioned behind the first areas **71A** and **71B** in the first direction **X**.

(358) As described above, external light entered the display panel **20** may be diffused and reflected to the outside of the display panel **20** by being transmitted through the first areas **71A** and **71B** of the first and third front covers **70A** and **70B**, or a part of the external light may be absorbed in the first areas **71A** and **71B** of the first and third front covers **70A** and **70B**. Accordingly, an amount of light arrived at the gap **G** may be reduced.

(359) In addition, light arrived at the gap **G** may be absorbed in the first and third side covers **90A** and **90B** positioned in the gap **G**, and accordingly, visibility of a boundary between the first display module **30A** and the third display module **30B** may be reduced.

(360) light reflected from the first and third side covers **90A** and **90B** and then traveling to the outside of the display panel **20** without being absorbed in the first and third side covers **90A** and **90B** may be diffused and reflected to the outside of the display panel **20** by being transmitted through the first areas **71A** and **71B** of the first and third front covers **70A** and **70B**, or a part of the light may be absorbed in the first areas **71A** and **71B**. Accordingly, an amount of light transmitted to the outside of the display panel **20** may be reduced, and thus, visibility of the boundary between the first display module **30A** and the second display module **30E**, caused by the gap **G**, may be reduced.

(361) The side cover **90** may be applied with a preset amount by a dispenser in a manufacturing process. The applied side cover **90** may be hardened by the subsequent operation. The side cover **90** may be formed of, for example, a nonconductive black resin.

(362) The side cover **90** may be applied to cover all of the rear surface of the front cover **70**, the side surface **45** of the substrate **40**, the chamfer portion **49** formed between the mounting surface **41** and the side surface **45**, and the chamfer portion **49** formed between the side surface **45** and the rear surface **43**.

(363) An operation of dispensing the side cover **90** may be performed on all of the four edges **E** of the substrate **40**. Accordingly, the side cover **90** may be dispensed to cover all the side surfaces **45** of the substrate **40**. Also, the entire of an area positioned outside the mounting surface **41** in the anisotropic conductive layer **47** may be covered by the side cover **90**.

(364) According to hardening of the side cover **90**, the side cover **90** may be bonded with the rear surface of the front cover **70** with respect to the first direction **X**, the side surface **45** of the substrate **40**, the chamfer portion **49** formed between the side surface **45** and the mounting surface **41**, and

the area positioned outside the mounting surface **41** in the anisotropic conductive layer **47**.

(365) The side cover **90** may include a photosensitive material. In this case, UV light, etc. may be irradiated to the side cover **90** in a subsequent operation to color the side cover **90** with a dark color. However, in a case in which the side cover **90** is formed of a translucent or opaque material without including a photosensitive material, the subsequent operation may be not needed.

(366) As described above, in the gap G between the first display module **30A** and the third display module **30B** in the third direction Z, the side end members **100** and **100'** of the first display module **30A** and the third display module **30B** may be positioned.

(367) Upon arraying of the first display module **30A** and the third display module **30B** in the third direction Z, the side end members **100** and **100'** of the first display module **30A** and the third display module **30B** may overlap with each other in the third direction Z to reduce a separation distance T between the first display module **30A** and the third display module **30B**.

(368) As described above, the side end member **100'** of the third display module **30B** being adjacent to the first display module **30A** may be positioned in a state of being rotated to 180 degrees in the first direction X with respect to the side end member **100** of the first display module **30A**.

(369) Accordingly, the plurality of mountains **111** and the plurality of valleys **112** of the side end member **100** of the first display module **30A** may be arranged alternately with the plurality of mountains **111** and the plurality of valleys **112** of the side end member **100'** of the third display module **30B** at the same height in the first direction X.

(370) Upon arraying of the first display module **30A** and the third display module **30B** in the third direction Z, the side end member **100** of the first display module **30A** may overlap with the side end member **100'** of the third display module **30B** in the third direction Z.

(371) That is, the plurality of mountains **111** of the side end member **100** of the first display module **30A** may be inserted into the grooves **113** formed by the plurality of valleys **112** of the side end member **100'** of the third display module **30B**.

(372) Accordingly, at least one portion of the side end member **100** of the first display module **30A** may overlap with at least one portion of the side end member **100'** of the third display module **30B** in the third direction Z to form a coupling section C, and a separation distance T between the first display module **30A** and the third display module **30B** may be reduced by a length of the overlapping portion of the coupling section C in the third direction Z.

(373) The side end members **100'** of the second display module **30E** and the third display module **30B** being adjacent to the first display module **30A** in the second direction Y and the third direction Z have been described above with reference to FIGS. 7 and 10. However, for example, a side end member **100** of the fourth display module **301** being opposite to the first display module **30A** with respect to the second display module **30E** may be positioned in a state of being rotated to 180 degrees in the first direction X with respect to the side end member **100'** of the second display module **30E**.

(374) Accordingly, the side end member **100** of the fourth display module **301** may be positioned in the first arrangement, like the first display module **30A**.

(375) Also, a side end member **100** of the fifth display module **30C** being opposite to the first display module **30A** with respect to the third display module **30B** may be positioned in a state of being rotated to 180 degrees in the first direction X with respect to the side end member **100'** of the third display module **30B**. Accordingly, the side end member **100** of the fifth display module **30C** may be positioned in the first arrangement, like the first display module **30A**.

(376) That is, the side end members **100** and **100'** of the neighboring display modules **30A** to **30P** may be intercrossed with a phase difference of 180 degrees. Accordingly, the side end members **100** and **100'** of the neighboring display modules **30A** to **30P** may overlap with each other in the second direction Y or the third direction Z, and thus, a substantial separation distance T between the display modules **30A** to **30P** may be reduced.

(377) Hereinafter, a side end member **200** according to another embodiment will be described. The side end member **200** which will be described below may have the same configuration as the side end member **100** described above except that the side end member **200** has a different shape for reducing a separation distance between the neighboring display modules **30A** to **30P**.

(378) For convenience of description, a side end member **200** of a first display module **30A** will be described, and a third display module **30B** arrayed to be adjacent to the first display module **30A** in the third direction **Z** will be described as an example.

(379) FIG. **11** is a cross-sectional view showing a portion of a display module of a display apparatus according to another embodiment, taken along the third direction, and FIG. **12** is a side view showing some components of the display module of the display apparatus of FIG. **11**, taken along the third direction.

(380) As shown in FIGS. **11** and **12**, the side end member **200** may cover at least one portion of the side cover **90** and extend along the second direction **Y**. In an example of the side end member **200** formed on the side surface **45** in the third direction **Z**, the side end portion **200** formed on the side surface **45** in the second direction **Y** may extend along the third direction **Z**.

(381) The side end member **200** may include a body **210** extending in one direction along the second direction **Y**, and a plurality of ribs **220** extending in the first direction **X** from the body **210** and arranged at intervals along the second direction **Y**.

(382) The body **210** may be formed at a rear portion of the side end member **200** in the first direction **X**, and the plurality of ribs **220** may extend from the body **210** toward the front direction **X**.

(383) The side cover **90** may include a side end portion **91** forming a side end **91e** of the side cover **90**, and a rear portion **92** extending from a rear end of the side end portion **91** in the first direction **X** and positioned at an inner location than the side end portion **91** in the third direction **Z**.

(384) In the side cover **90**, the side end portion **91** may be formed in a front portion of the side cover **90** in the first direction **X**, and the rear portion **92** may be formed in a rear portion of the side cover **90** in the first direction **X**.

(385) The rear portion **92** may cover a rear portion of the side surface **45** in the first direction **X**, and extend to an inner location in the third direction **Z** along the side surface **45** and the chamfer portion **49** between the side surface **45** and the rear surface **43**. Accordingly, the rear portion **92** may be positioned at an inner location than the side end portion **91** in the third direction **Z** with respect to the side end portion **91**.

(386) A portion of the side end member **200**, positioned on the side end portion **91** in the third direction **Z**, is defined as a first portion **201**, and a portion of the side end member **200**, positioned on the rear portion **92** in the third direction **Z**, is defined as a second portion **202**. In this case, the body **210** may be formed in the second portion **202**.

(387) That is, the body **210** may extend along the second direction **Y** on the rear portion **92** in the third direction **Z**.

(388) The plurality of ribs **220** may be formed in the first portion **201**. Also, the plurality of ribs **220** may be formed across the first portion **201** and the second portion **202**.

(389) That is, the plurality of ribs **220** may be arranged at intervals in the second direction **Y** on the side end portion **91** in the third direction **Z**, and at least one portions of the plurality of ribs **220** may be formed on the rear portion **92** in the third direction **Z**.

(390) As described above, because the plurality of ribs **220** extend from the body **210** in the first direction **X**, the plurality of ribs **220** may be positioned on the side end portion **91** in the third direction **Z**, and at least one portion of the plurality of ribs **220** may be positioned on the rear portion **92** in the third direction **Z**, like the body **210**, although the embodiment is not limited thereto.

(391) In an embodiment, the plurality of ribs **220** may be positioned only at a location corresponding to the side end portion **91** in the third direction **Z**, and the body **210** may be

positioned only at a location corresponding to the rear portion **92** in the third direction Z.

(392) Each of the plurality of ribs **220** may include a front end portion **221** positioned on the side end portion **91** in the third direction Z and forming a front end of the rib **220** in the first direction X, and a rear end portion **222** positioned on the rear portion **92** in the third direction Z, forming a rear end of the rib **220** in the first direction X, and connected with the body **210**.

(393) Because the front end **221** of each of the plurality of ribs **220** is positioned on the side end portion **91** and the rear end **222** is positioned on the rear portion **92**, the plurality of ribs **220** may extend in the first direction X to be positioned on the rear portion **92** through the side end portion **91**.

(394) The body **210** may overlap with the metal plate **60** in the first direction X and be in contact with the metal plate **60**. Accordingly, charges generated by an electrostatic discharge and then entered the side cover **90** may be guided to the metal plate **60** through the side end member **200**.

(395) A rear end of the body **210** in the first direction X may be in contact with the metal plate **60**, and a front end of the body **210** in the first direction X may be positioned on the rear portion **92** in the third direction Z.

(396) Because the plurality of ribs **220** are arranged at intervals in the second direction Y, as described above, the side end member **200** may include a plurality of spaces **230** formed between the plurality of ribs **220**.

(397) The plurality of spaces **230** may be spaces in which a plurality of ribs **220** formed on the side end portion **91** of the third display module **30B** are inserted, which will be described below.

(398) Accordingly, each of the plurality of spaces **230** may have a width that is equal to or greater than a width of each of the plurality of ribs **220** in the second direction Y such that the plurality of ribs **220** formed on the side end portion **91** of the third display module **30B** are inserted in the plurality of spaces **230**.

(399) Hereinafter, an arrangement structure for reducing a separation distance between the display modules **30A** to **30P** will be described in detail.

(400) FIG. **13** schematically shows a process in which display modules of the display apparatus shown in FIG. **11** are arrayed, FIG. **14** schematically shows a process in which display modules of an embodiment are arrayed, and FIG. **15** is a cross-sectional view showing some components of the embodiment of a display module of FIG. **11**, taken along the third direction.

(401) As shown in FIGS. **13** and **14**, because a plurality of ribs **220** are positioned at locations corresponding to the side end portion **91** in the third direction Z, the plurality of ribs **220** positioned on the side end portion **91** in the third direction Z may be protrusions arranged at intervals along the second direction Y.

(402) As described above, a plurality of ribs **220** of the first display module **30A** may be inserted into a plurality of spaces **230** of the third display module **30B** in the third direction Z.

(403) Accordingly, as shown in FIG. **15**, a separation distance T between the first display module **30A** and the third display module **30B** may be equal to a thickness t of the side end member **200** in the third direction Z, and thus, the separation distance T between the first display module **30A** and the third display module **30B** may be reduced, resulting in a reduction of visibility of a seam between the first display module **30A** and the third display module **30B**.

(404) As shown in FIG. **13**, the side end member **200** of the first display module **30A** may be positioned on the side cover **90** such that a first rib **225** is positioned at a front end in the second direction Y and a space (which may be referred to as a hindmost space) **235** is positioned at a rear end in the first direction X.

(405) A side end member **200'** of the third display module **30B** may be positioned on the side cover **90** such that a space (which may be referred to as a foremost space) **235'** is positioned at a front end in the first direction X and a first rib **225'** is positioned at a rear end in the first direction X, reversely from the side end portion **200** of the first display module **30A**.

(406) A side end member such as the side end member **200** of the first display module **30A** in

which the first rib **225** is positioned at the front end in the second direction Y is defined as a side end member **200** of a first arrangement, and a side end member such as the side end member **200'** of the third display module **30B** in which the first rib **225'** is positioned at the rear end in the first direction X is defined as a side end member **200'** of a second arrangement.

(407) In a case in which the first display module **30A** is positioned to face the third display module **30B** in the third direction Z, the plurality of ribs **220** of the first display module **30A** may be positioned to correspond to a plurality of spaces **230'** of the third display module **30B** in the third direction Z. Accordingly, a front end **45e** of the first display module **30A** in the second direction Y and a front end **45e** of the third display module **30B** in the second direction Y may be positioned at the same height in the second direction Y such that the plurality of ribs **220** of the first display module **30A** are inserted into the plurality of spaces **230'** of the third display module **30B**.

(408) The front end **45e** of the first display module **30A** in the second direction Y and the front end **45e** of the third display module **30B** in the second direction Y may be formed as the side surface **45** of the substrate **40** or the side cover **90**. For convenience of description, the front ends **45e** are denoted by solid lines in FIGS. **13** and **14**.

(409) That is, when the side end member **200** of the first display module **30A** and the side end member **200'** of the third display module **30B** are positioned in a state rotated by 180 degrees with respect to each other, the plurality of ribs **220** of the first display module **30A** may be positioned to correspond to the plurality of spaces **230'** of the third display module **30B** on the same line in the third direction Z, and accordingly, the first display module **30A** may be arrayed to correspond to the third display module **30B** on the same line with respect to the first direction X.

(410) As described above, the side end member **200'** of the third display module **30B** being adjacent to the first display module **30A** may be positioned in a state of being rotated to 180 degrees in the first direction X with respect to the arrangement of the side end member **200** of the first display module **30A**.

(411) As shown in FIG. **14**, each of the side end member **200** of the first display module **30A** and the side end member **200'** of the third display module **30B** may be positioned in the first arrangement on the side cover **90** such that the side end member **200** of the first display module **30A** is symmetrical to the side end member **200'** of the third display module **30B** in the third direction Z.

(412) In this case, in the first display module **30A** and the third display module **30B** arranged in the third direction Z, the plurality of ribs **220** of the first display module **30A** may be positioned to correspond to the plurality of ribs **220** of the third display module **30B** on the same line in the third direction Z. Accordingly, in a case in which the first display module **30A** is arrayed with the third display module **30B** by positioning the front end **45e** of the first display module **30A** and the front end **45e** of the third display module **30B** at the same height in the second direction Y, the plurality of ribs **220** of the first display module **30A** may be not inserted into the plurality of spaces **230** of the third display module **30B**.

(413) That is, in the case in which the side end member **200** of the first display module **30A** is symmetrical to the side end member **200'** of the third display module **30B** in the third direction Z, the plurality of ribs **220** of the first display module **30A** may be not positioned to correspond to the plurality of spaces **230** of the third display module **30B** on the same line in the third direction Z. Therefore, the first display module **30A** may be not arrayed to correspond to the third display module **30B** on the same line with respect to the first direction X.

(414) Accordingly, the first display module **30A** may be arrayed with the third display module **30B** in the third direction Z by intercrossing the front end **45e** of the first display module **30A** in the second direction Y with the front end **45e** of the third display module **30B** in the second direction Y, in the first direction X, instead of positioning the front end **45e** of the first display module **30A** and the front end **45e** of the third display module **30B** on the same line.

(415) In this case, the first display module **30A** and the third display module **30B** may be arrayed in

the third direction Z by positioning the third display module **30B** at a higher location in the first direction X than the first display module **30A** such that the first rib **225** of the first display module **30A** is inserted into a first space **236** formed behind the first rib **225** of the third display module **30B** in the second direction Y.

(416) In example embodiments, any of the plurality of display modules **30A** to **30P** may have either or both of the side end members **200** and **200'**. For example, in another embodiment, instead of the first display module **30A** including a side end module **200** and the third display module **30B** including a side end module **200'**, they may be reversed such that the first display module **30A** includes a side end module **200'** and the second display module **30E** includes a side end module **200**. In other example embodiments, any of the plurality of display modules **30A** to **30P** may have side end members **200** at one side and a side end member **200'** at another side.

(417) In the display apparatus according to an embodiment, each display module may be sealed by the front cover in the front direction, by the side cover at the side surface, and by the metal plate at the rear side, and by the side end member additionally positioned on the side surface and grounded to the metal plate, ESD internal pressure against an electrostatic discharge that may occur in the display module in a process of manufacturing and transferring the display module and after the display module is assembled into the display apparatus may be improved.

(418) The display apparatus according to an embodiment may provide a seamless effect by which a seam between the display modules is invisible by improving a shape of the side end member of each display module to reduce a gap between the neighboring display modules.

(419) While various embodiments have been particularly shown and described, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

Claims

1. A display module comprising: a substrate having a mounting surface on which a plurality of inorganic light-emitting diodes emitting light in a first direction are mounted, a side surface, and a rear surface being opposite to the mounting surface; a front cover bonded with the mounting surface of the substrate and covering the mounting surface in the first direction; a metal plate bonded with the rear surface of the substrate; a side cover surrounding the side surface of the substrate; and a side end member covering at least one portion of the side cover, and extending along the side surface of the substrate in a second direction that is orthogonal to the first direction, wherein the side end member comprises a body extending in the second direction, and a plurality of ribs extending from the body in the first direction and arranged at intervals along the second direction.
2. The display module of claim 1, wherein the side cover comprises: a side end portion forming a side end of the side cover, and a rear portion extending in a rear direction from the side end portion with respect to the first direction and positioned at an inner location than the side end portion in a third direction that is orthogonal to the first direction and the second direction, and wherein the body is provided on the rear portion in the third direction.
3. The display module of claim 2, wherein the plurality of ribs are provided on the side end portion.
4. The display module of claim 2, wherein the plurality of ribs are provided on the side end portion in the third direction and the rear portion in the third direction.
5. The display module of claim 4, wherein the body is formed at rear ends of the plurality of ribs.
6. The display module of claim 1, wherein a length of each of the plurality of ribs in the second direction is equal to or less than a length of a separation distance between the plurality of ribs in the second direction.
7. The display module of claim 1, wherein a conductivity of the side end member is greater than a conductivity of the side cover.

8. The display module of claim 1, wherein a color of the side end member is a black color.
 9. The display module of claim 1, wherein the side end member comprises a metal material.
 10. A display module comprising: a substrate having a mounting surface on which a plurality of inorganic light-emitting diodes are mounted, a side surface, and a rear surface being opposite to the mounting surface; a front cover bonded with the mounting surface of the substrate and covering the mounting surface of the substrate; a metal plate bonded with the rear surface of the substrate; a side cover surrounding the side surface of the substrate; and a side end member covering at least one portion of the side cover and at least one portion of the metal plate, the side end member comprising a first end portion, a second end portion spaced from the first end portion in a direction in which the mounting surface faces, and a side end portion extending from the first end portion to the second end portion and forming a side end; wherein the first end portion of the side end member covers at least one portion of the side end of the side cover, wherein the second end portion of the side end member contacts the metal plate and is grounded to the metal plate, and wherein the side end portion comprises mountains and valleys alternately arranged in the direction in which the mounting surface faces.
 11. The display module of claim 10, wherein a mountain from among the mountains is provided at the first end portion, and wherein a valley among the valleys is provided at the second end portion.
 12. The display module of claim 10, wherein a conductivity of the side end member is greater than a conductivity of the side cover.
 13. The display module of claim 10, wherein a height of each of the mountains and a depth of each of the valleys are the same.
 14. The display module of claim 13, wherein the mountains and the valleys have a same volume.
 15. A display apparatus comprising: a display module array comprising a plurality of display modules arranged in a matrix, wherein each of the plurality of display modules comprises: a substrate having a mounting surface on which a plurality of inorganic light-emitting diodes in a first direction are mounted, a side surface, and a rear surface being opposite to the mounting surface; a front cover bonded with the mounting surface of the substrate and covering the mounting surface of the substrate; a metal plate bonded with the rear surface of the substrate; a side cover surrounding the side surface of the substrate; and a side end member covering at least one portion of the side cover and extending along the side surface of the substrate in a second direction that is orthogonal to the first direction, wherein the side end member comprises a body extending in the second direction, and a plurality of ribs extending from the body in the first direction and arranged at intervals along the second direction.
 16. The display apparatus of claim 15, wherein the plurality of display modules comprises a first display module and a second display module, and wherein the side end member of the first display module engages the side end member of the second display module.
 17. The display apparatus of claim 16, wherein a length of each of the plurality of ribs in the second direction is equal to or less than a length of a separation distance between the plurality of ribs in the second direction.
 18. The display apparatus of claim 16, wherein the side cover comprises: a side end portion forming a side end of the side cover, and a rear portion extending in a rear direction from the side end portion with respect to the first direction and in a third direction that is orthogonal to the first direction and the second direction, and wherein the body is provided on the rear portion.
 19. The display apparatus of claim 18, wherein the plurality of ribs are provided on the side end portion.
 20. The display apparatus of claim 18, wherein the plurality of ribs are provided on the side end portion in the third direction and the rear portion in the third direction.
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